

Continental-Scale Water Cycle Modeling:

Integrating Climate, Surface Water, Groundwater, Sea-Level, and Human Use

Project Goal:

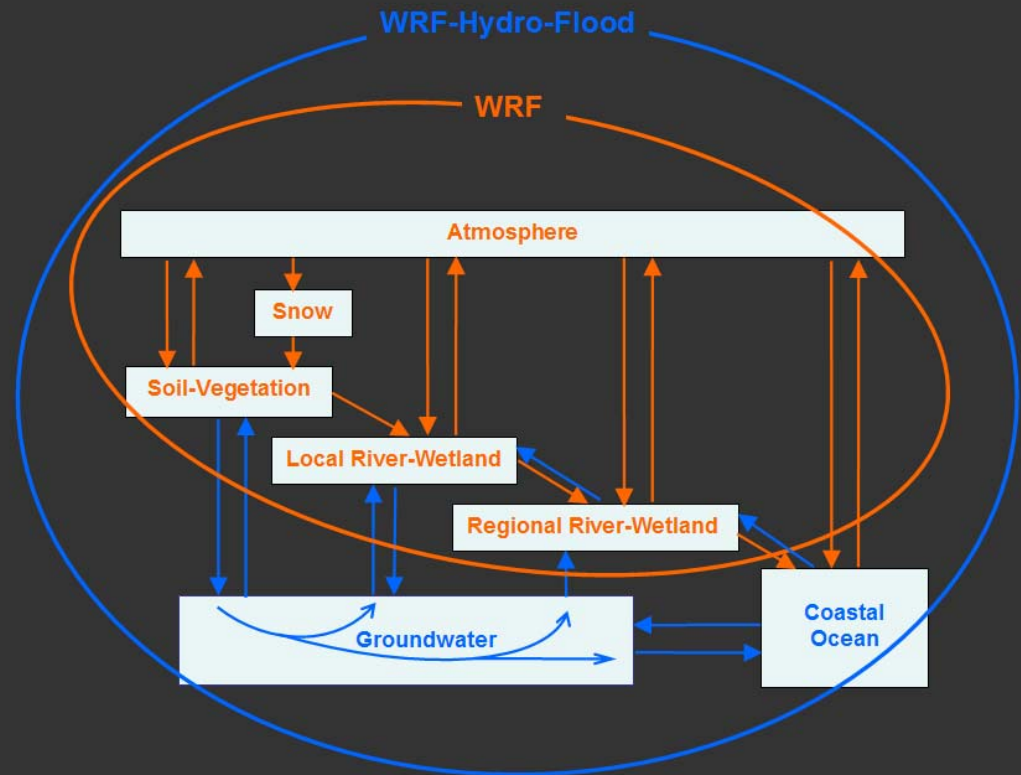
- Assess US freshwater resource climate variability & change increasing human demands

Scale:

- 5km - Continental

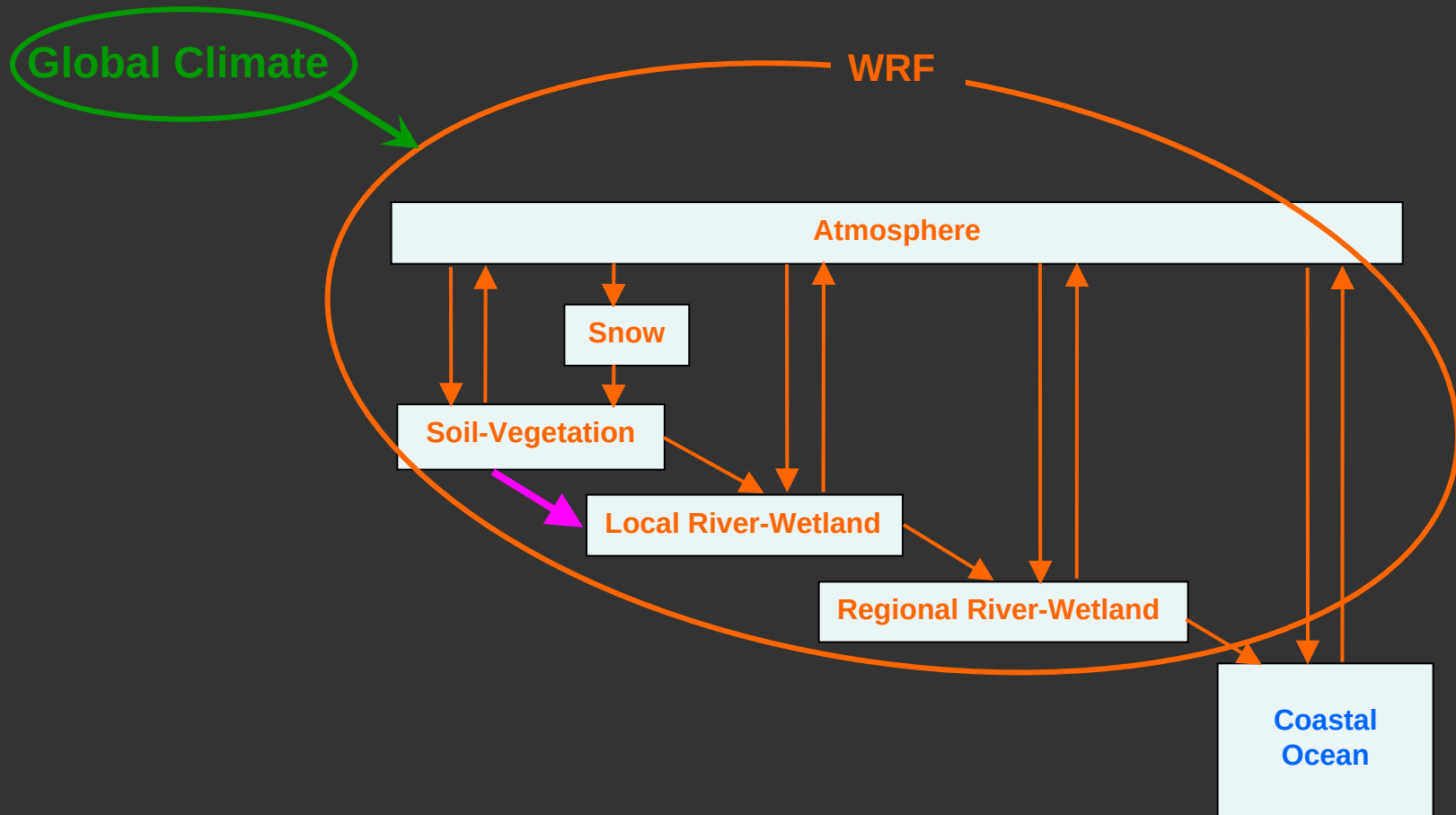
Tool:

- Integrated Water Cycle Model, **WRF-Hydro-Flood** groundwater and interfaces



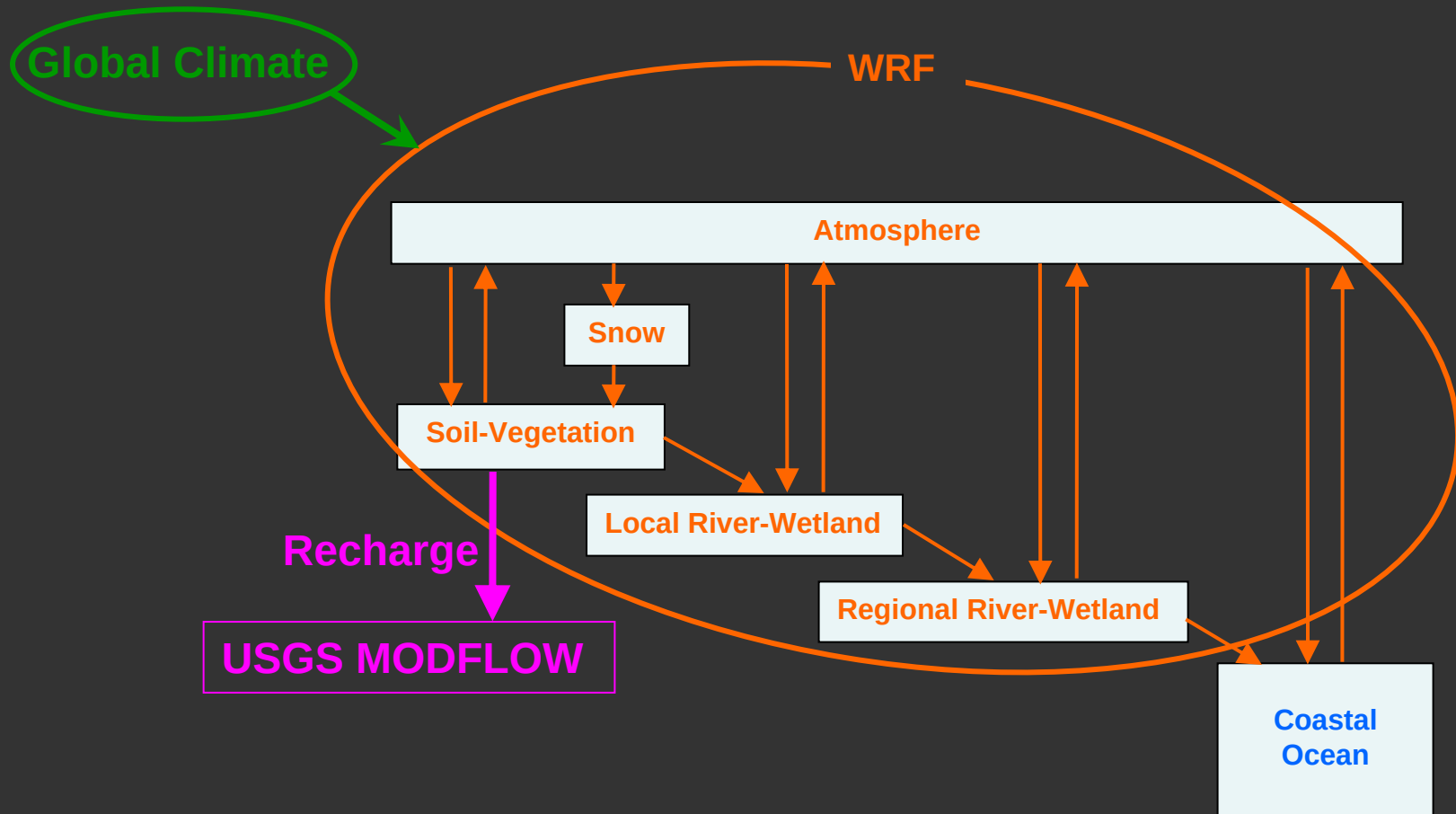
Typical Approach:

- Regional Climate Model Downscaling



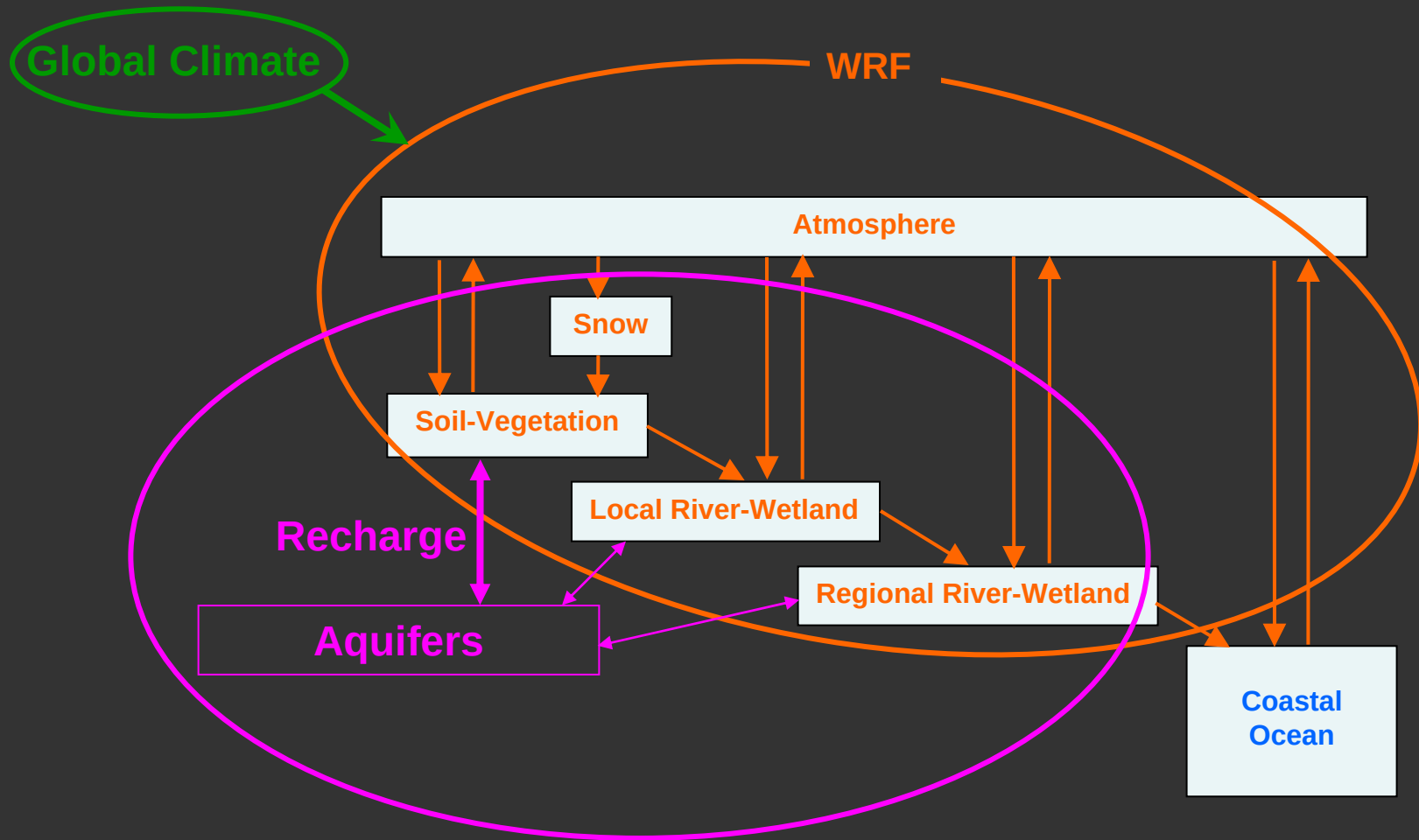
Typical Approach:

- Regional Climate Model Downscaling



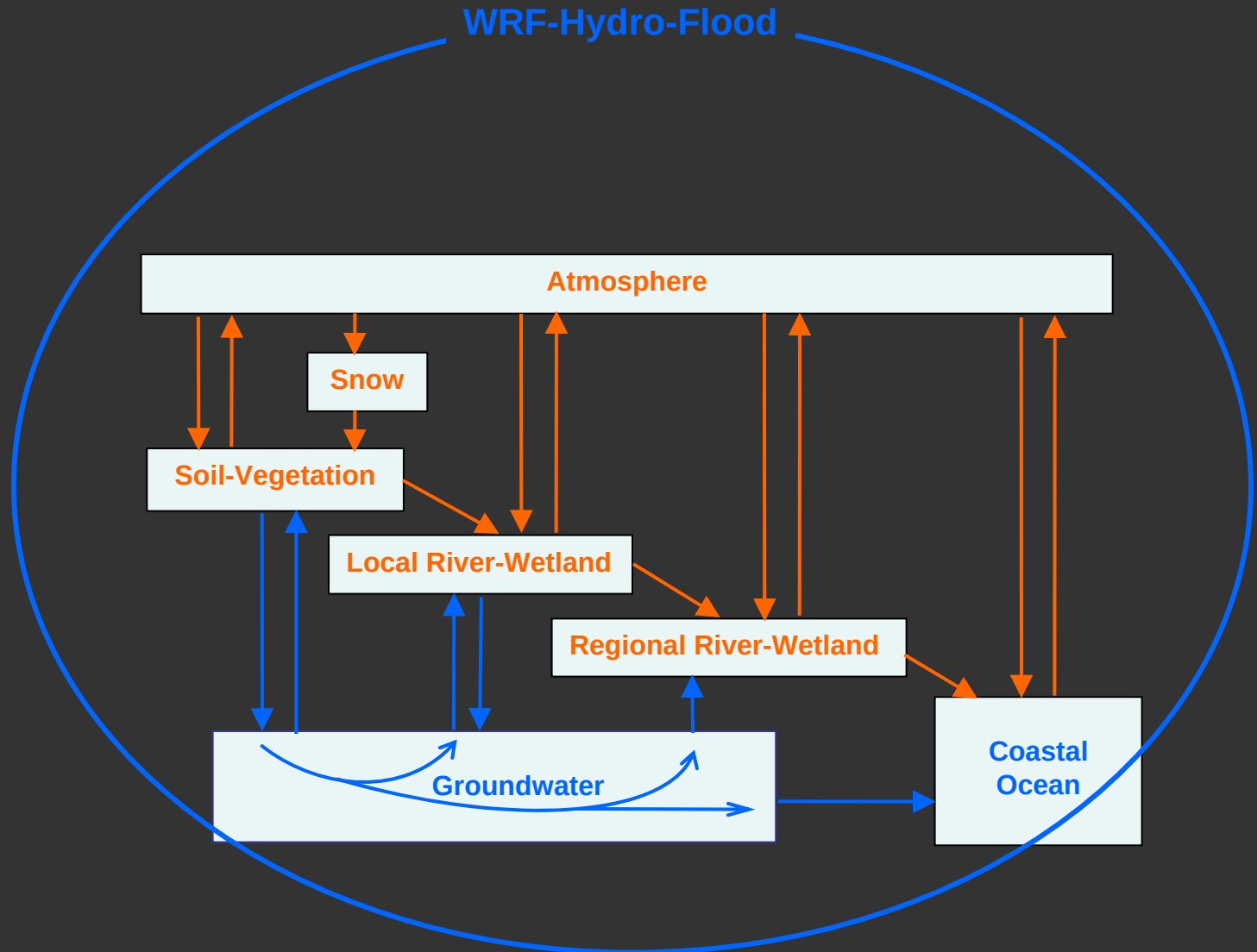
Typical Approach:

- Regional Climate Model Downscaling



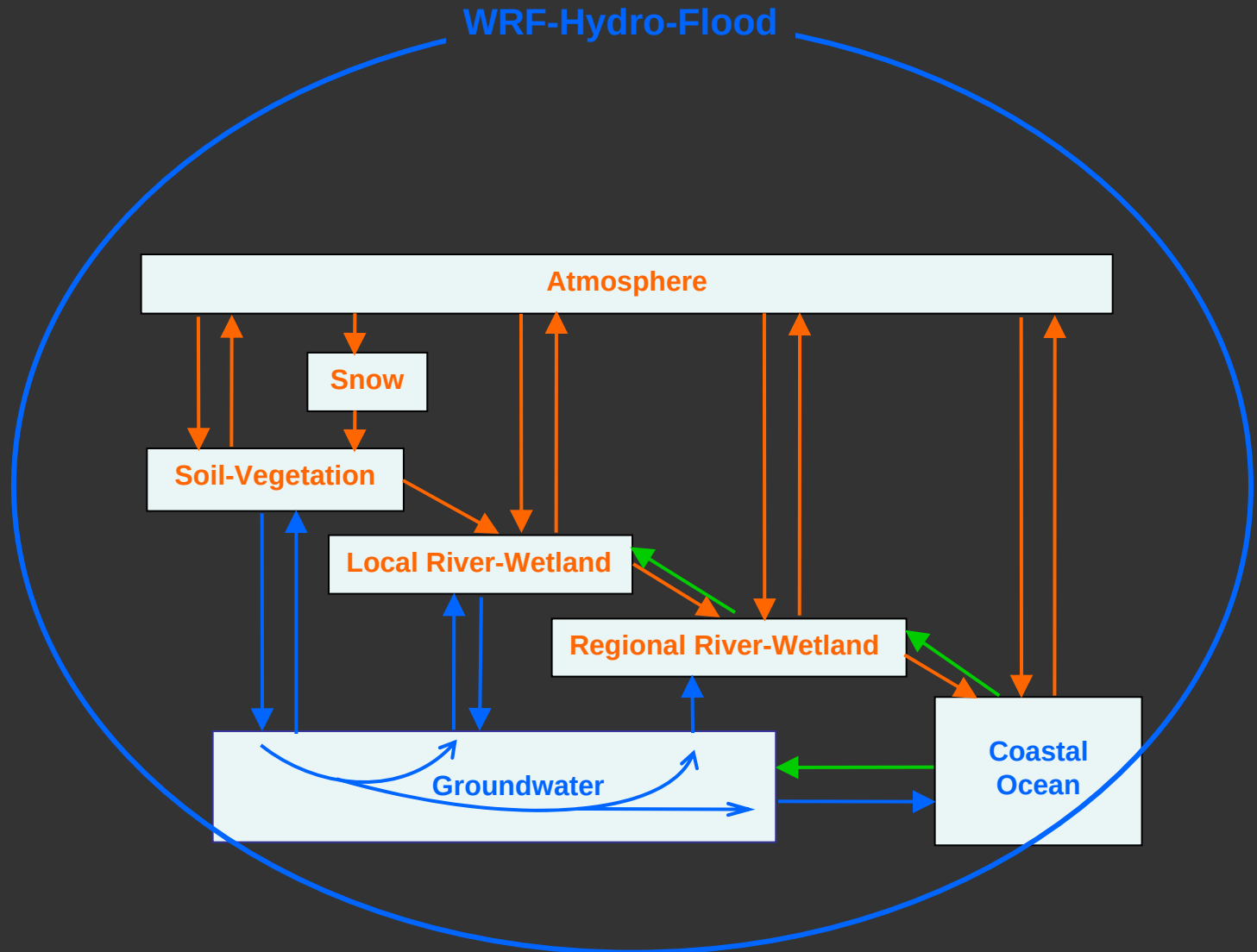
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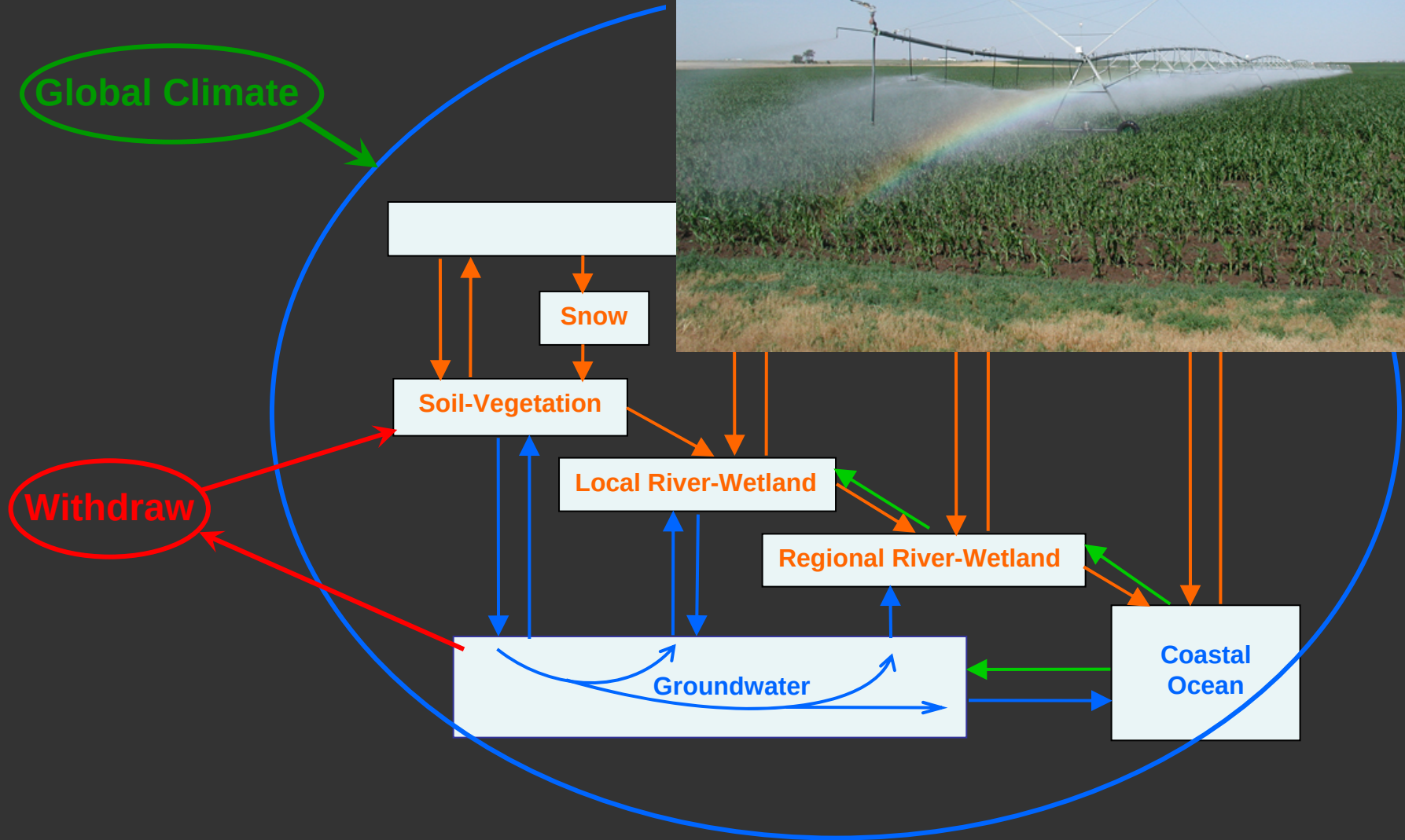
Typical Approach:

- Regional Climate Model Downscaling



Typical Approach:

- Regional Climate Model Downscale



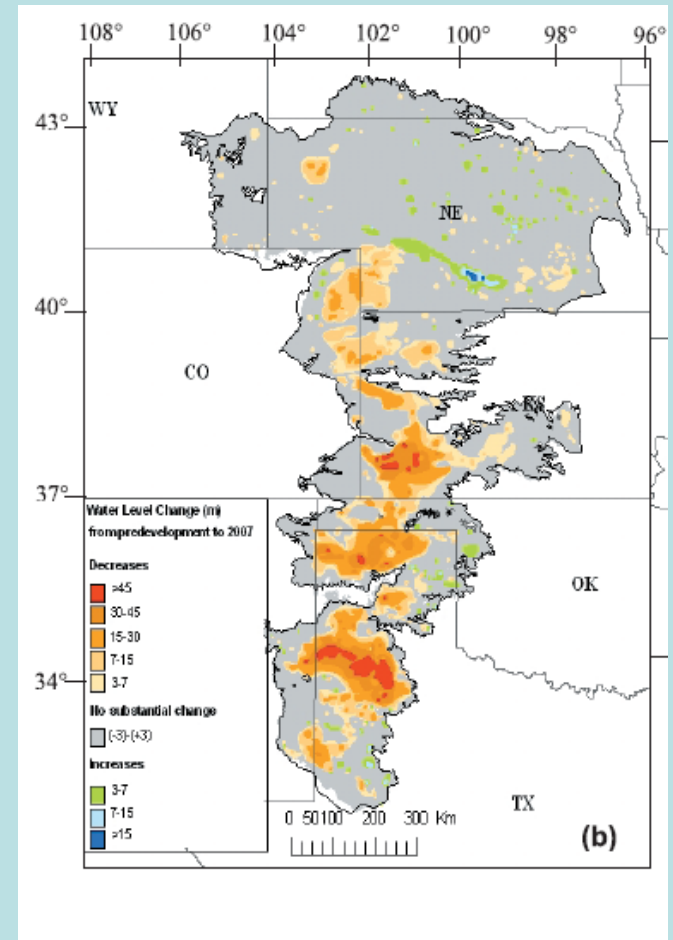
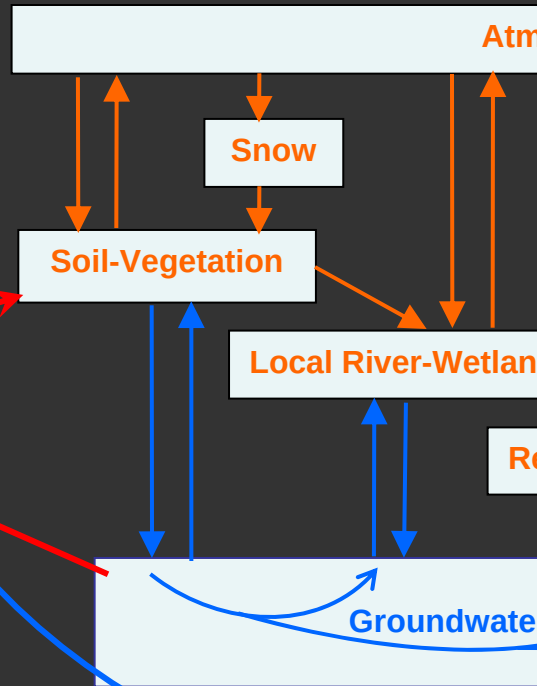
Typical Approach:

- Regional Climate Model Downscaling

WRF-Hy

Global Climate

Withdraw



Typical Approach:

- Regional Climate Model Downscaling

WRF-Hydro

Global Climate

Atmos

Snow

Soil-Vegetation

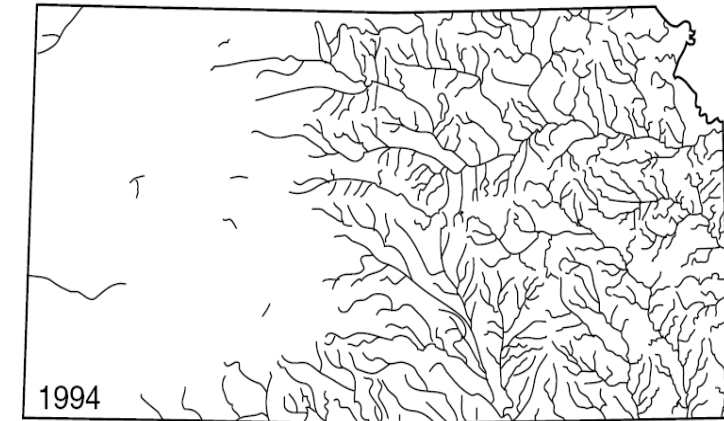
Local River-Wetland

Regional River-Wetland

Groundwater

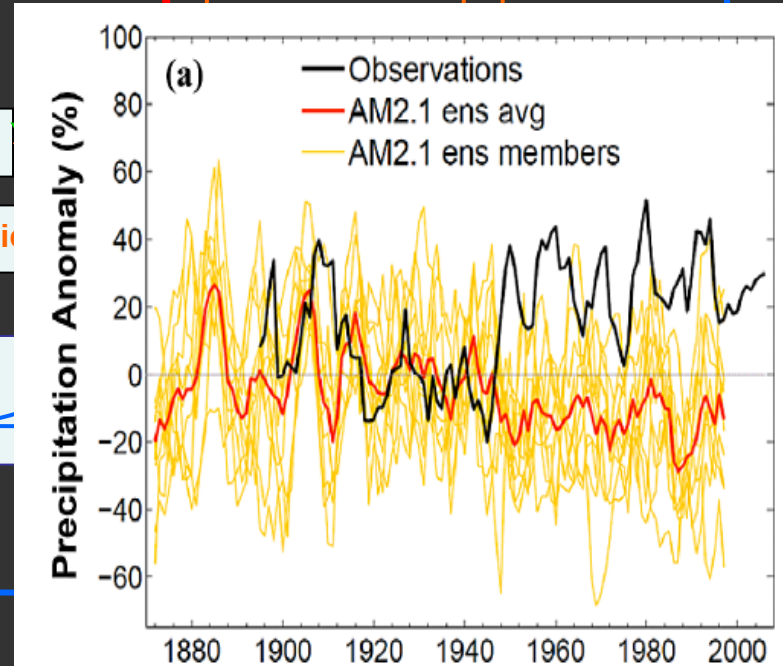
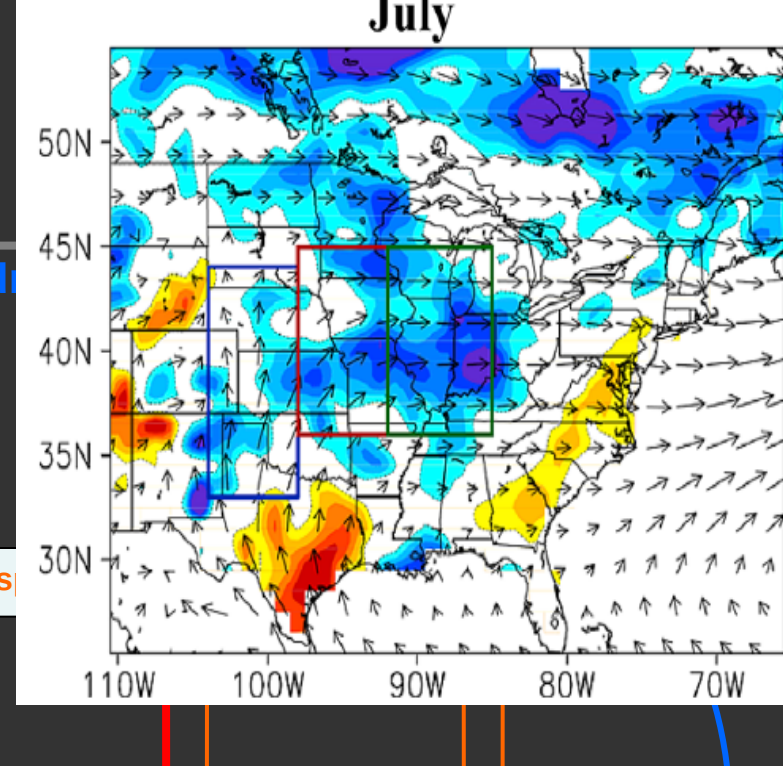
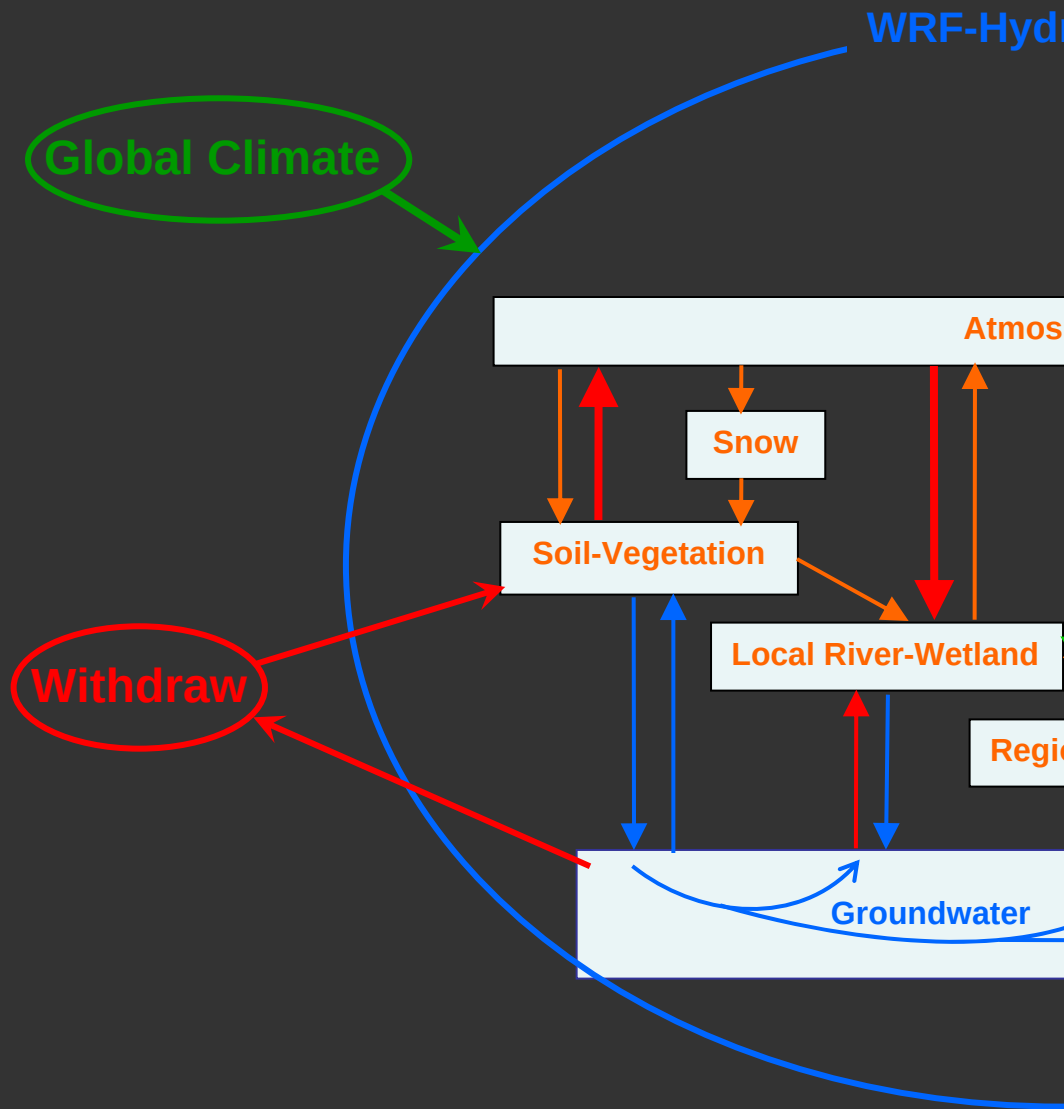
Coastal
Ocean

Withdraw



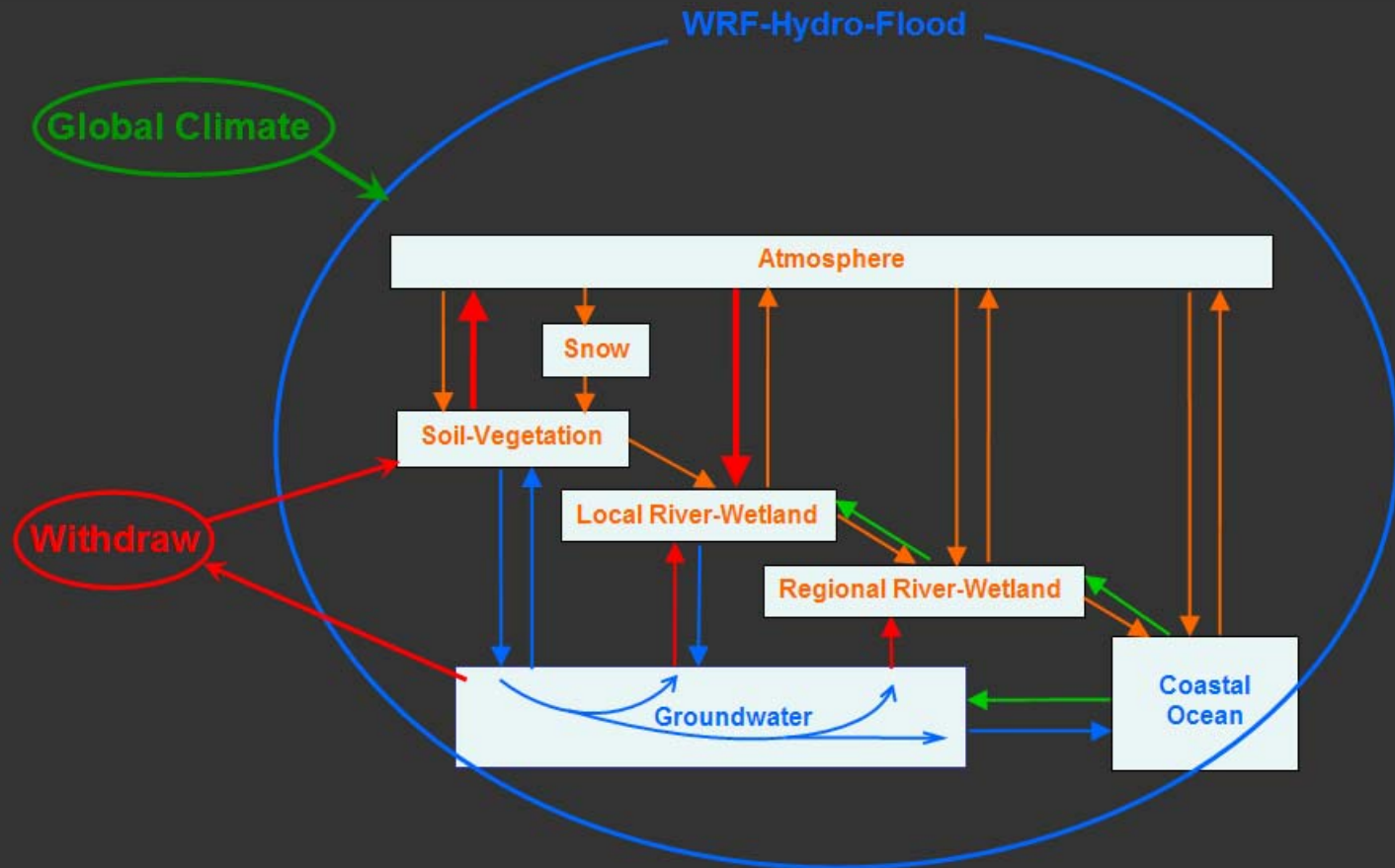
Typical Approach:

- Regional Climate Model Downscaling

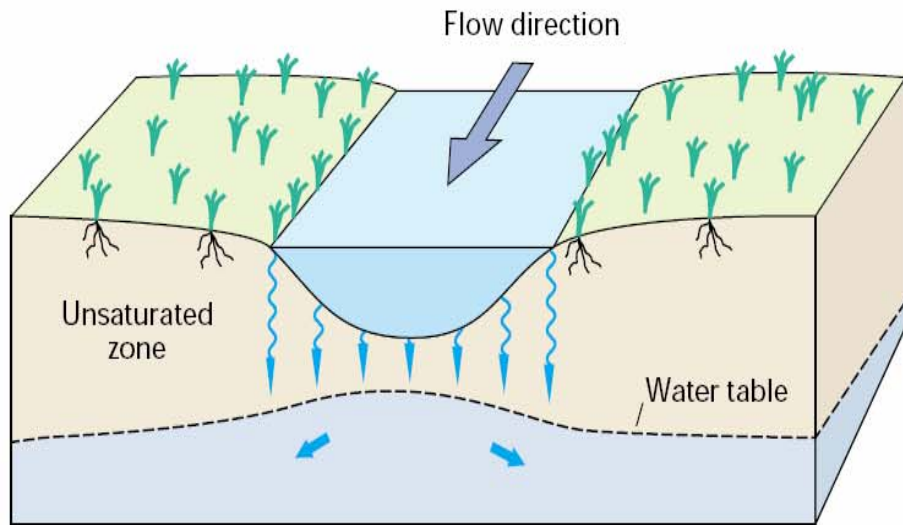


Advantages:

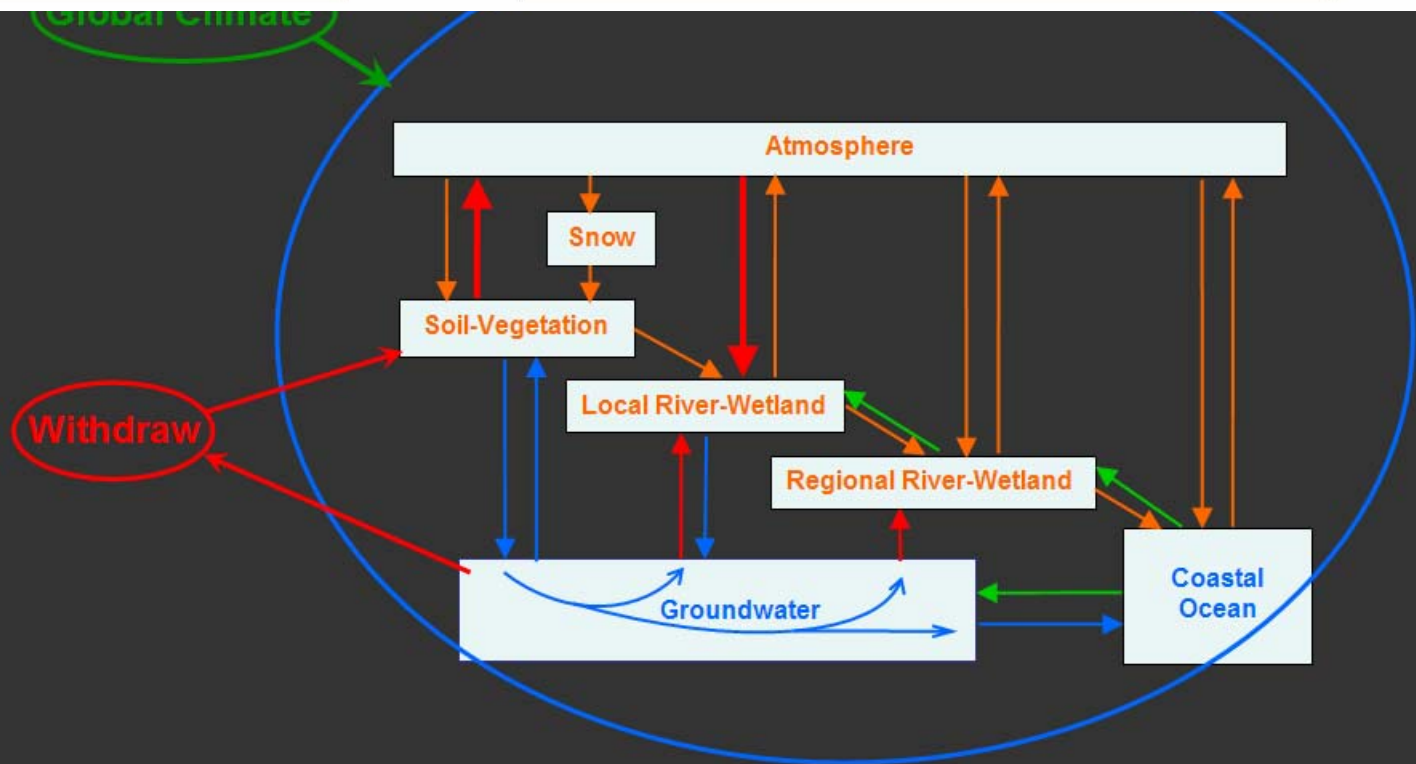
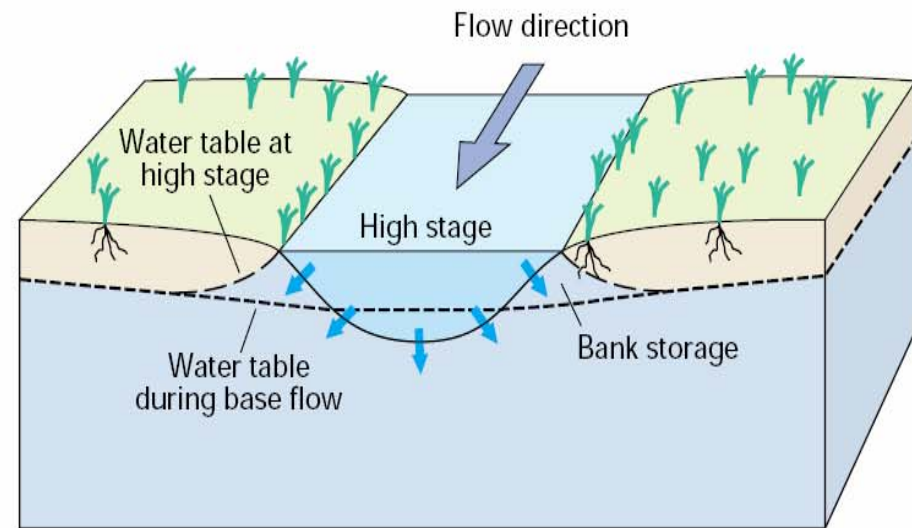
- Fully coupled, internally consistent water cycle
- Enabled several interfaces related to the GW
- Takes into account of GW buffer
- Physical framework for building in biogeochemistry



DISCONNECTED STREAM

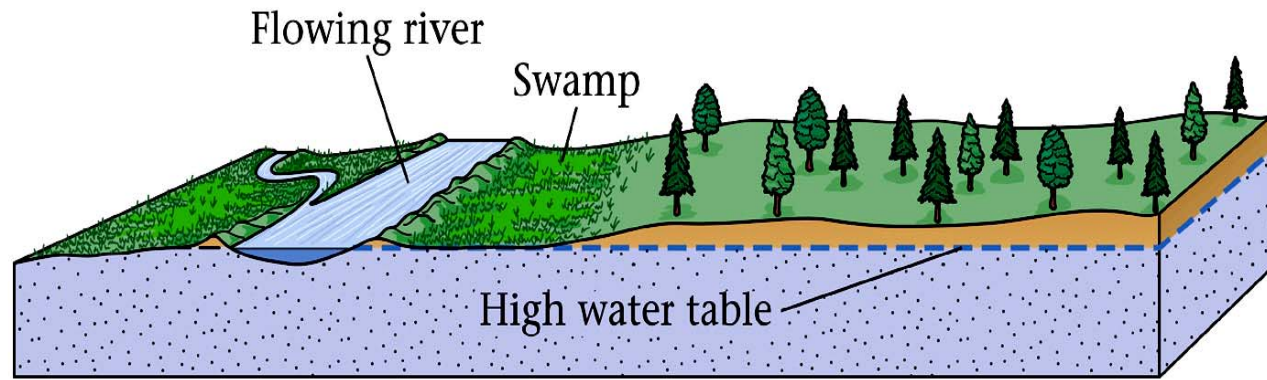


BANK STORAGE

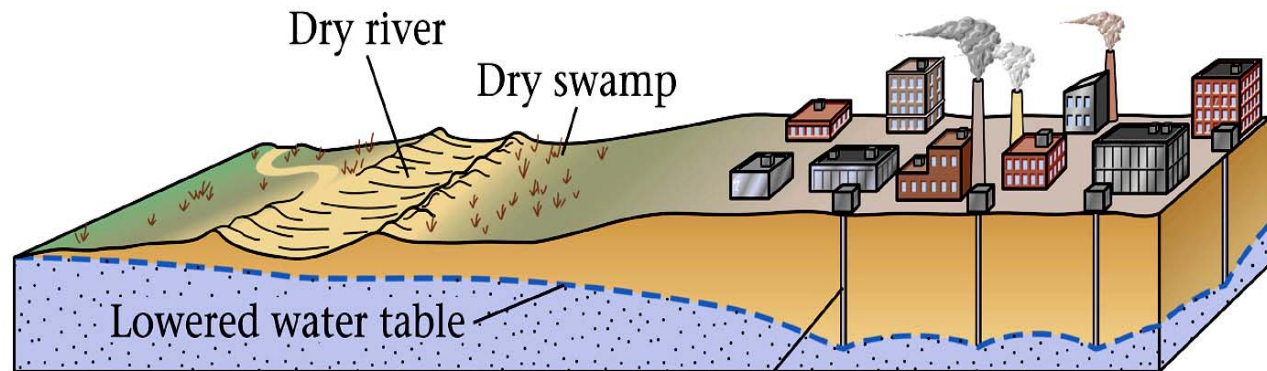


Advantages:

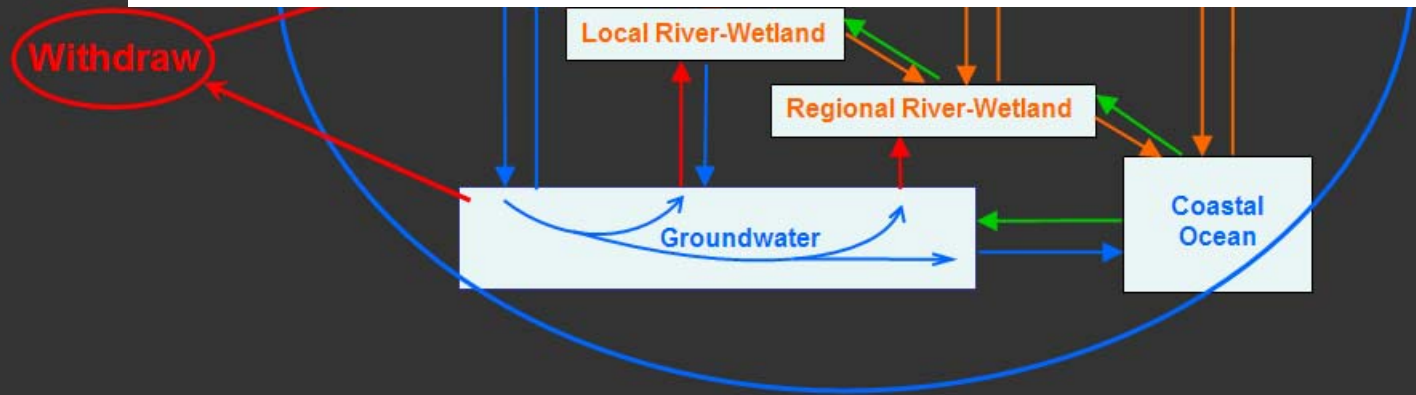
- Fully coupled, internal
- Enabled several inter
- Takes into account o
- Physical framework



(a)

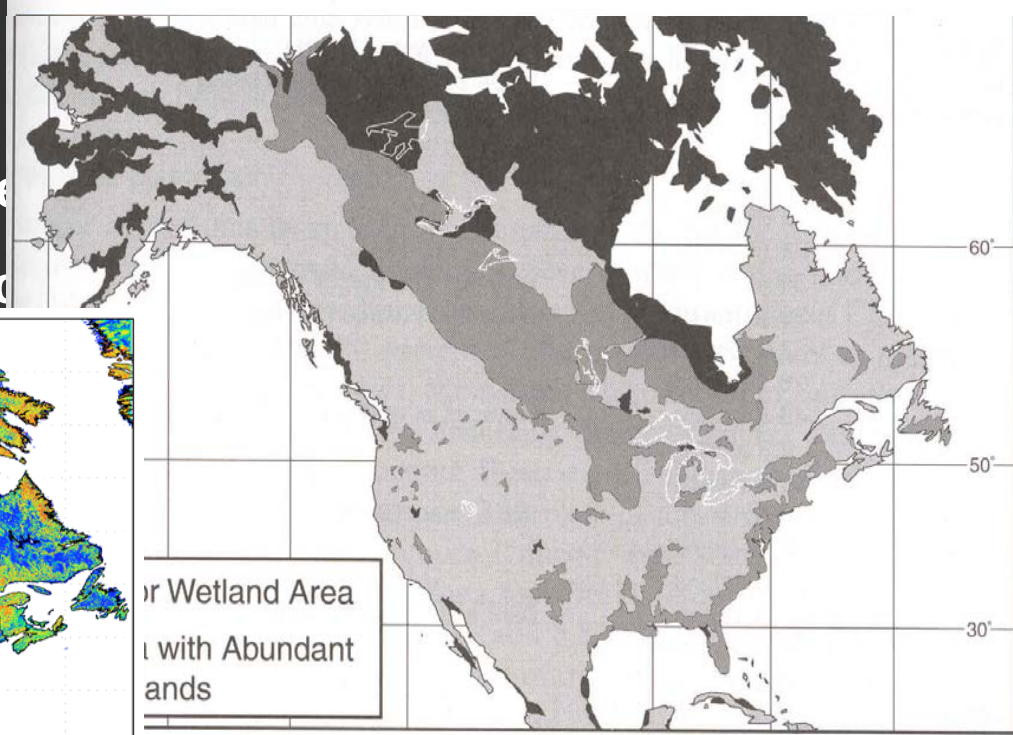
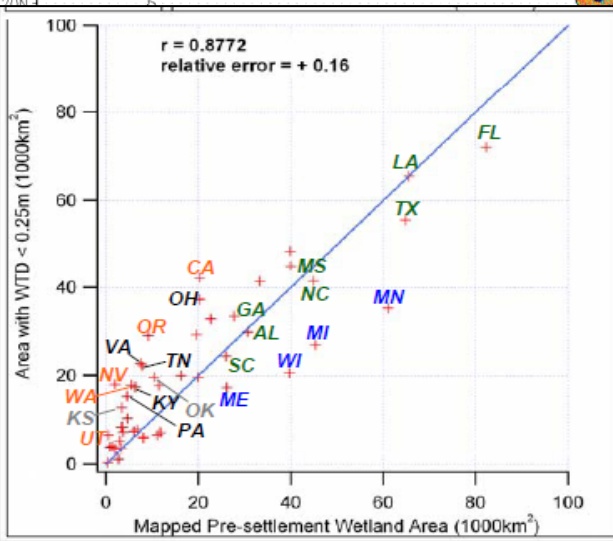
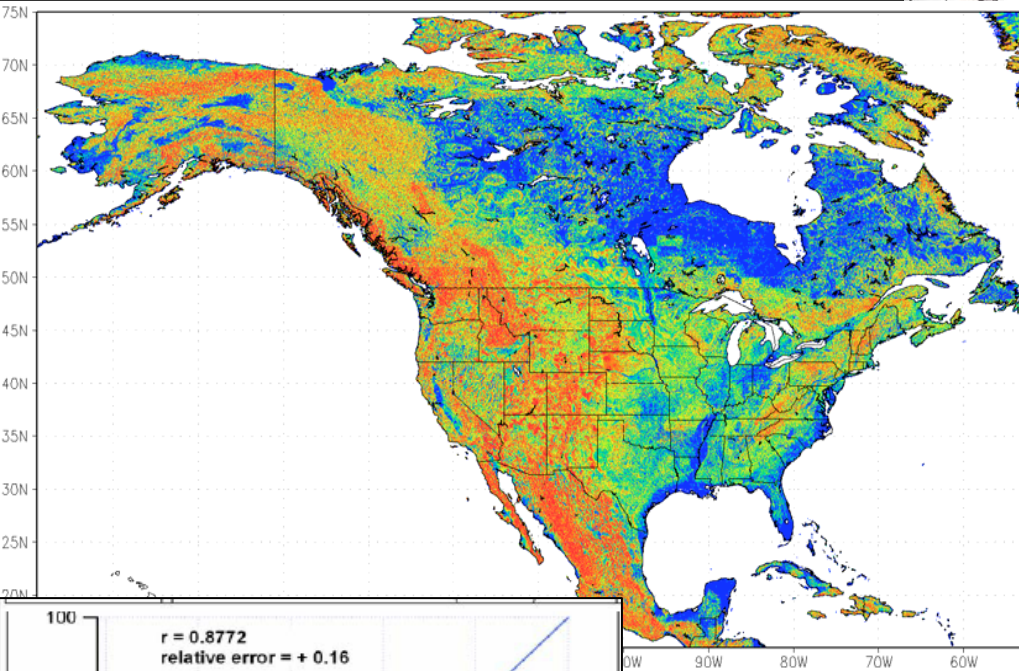


(b)

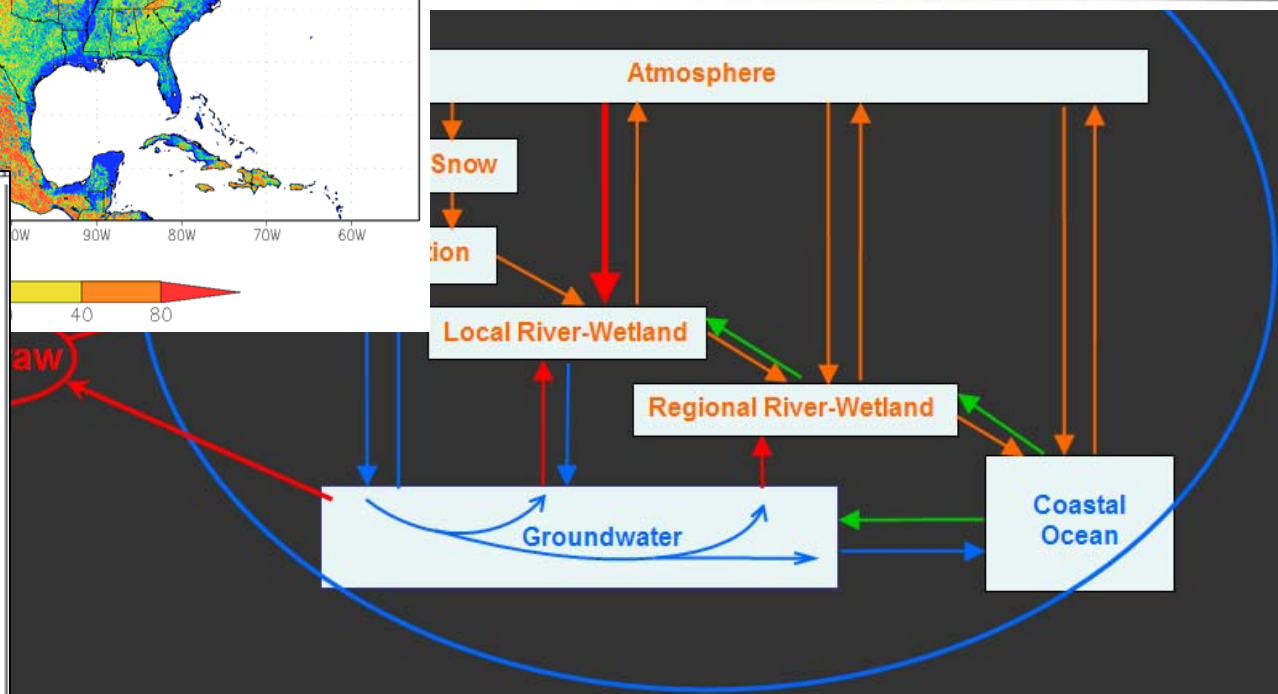


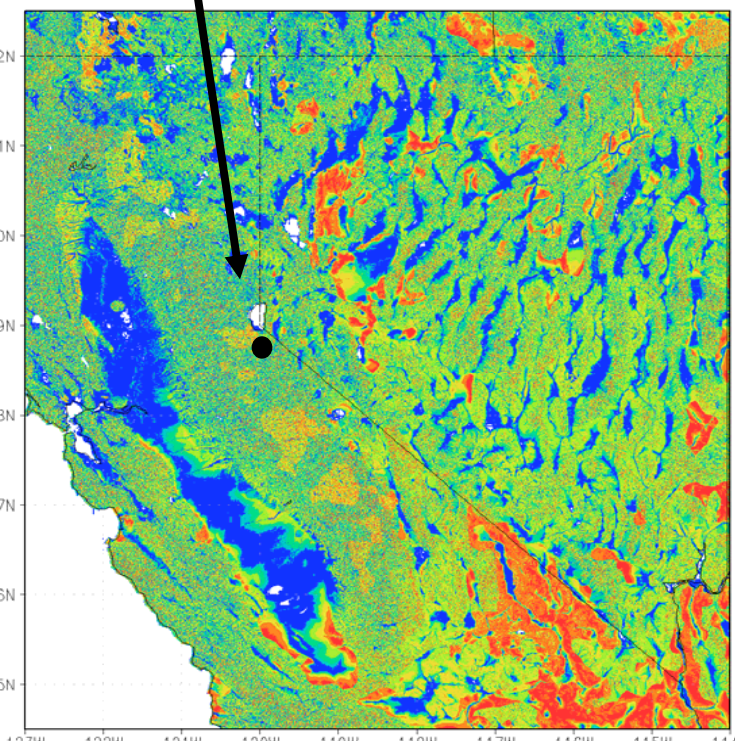
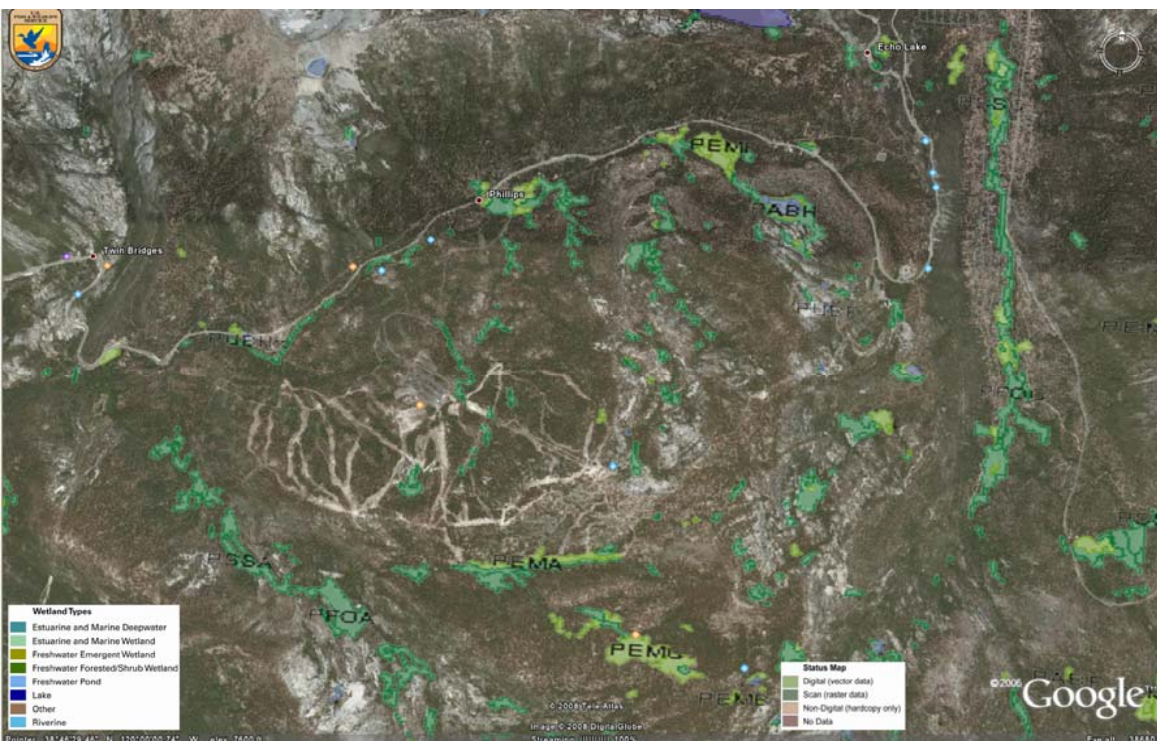
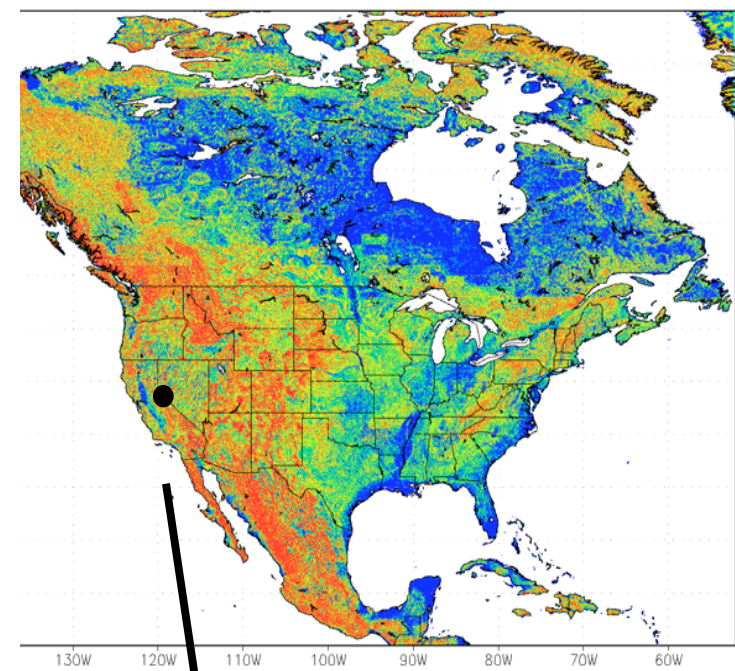
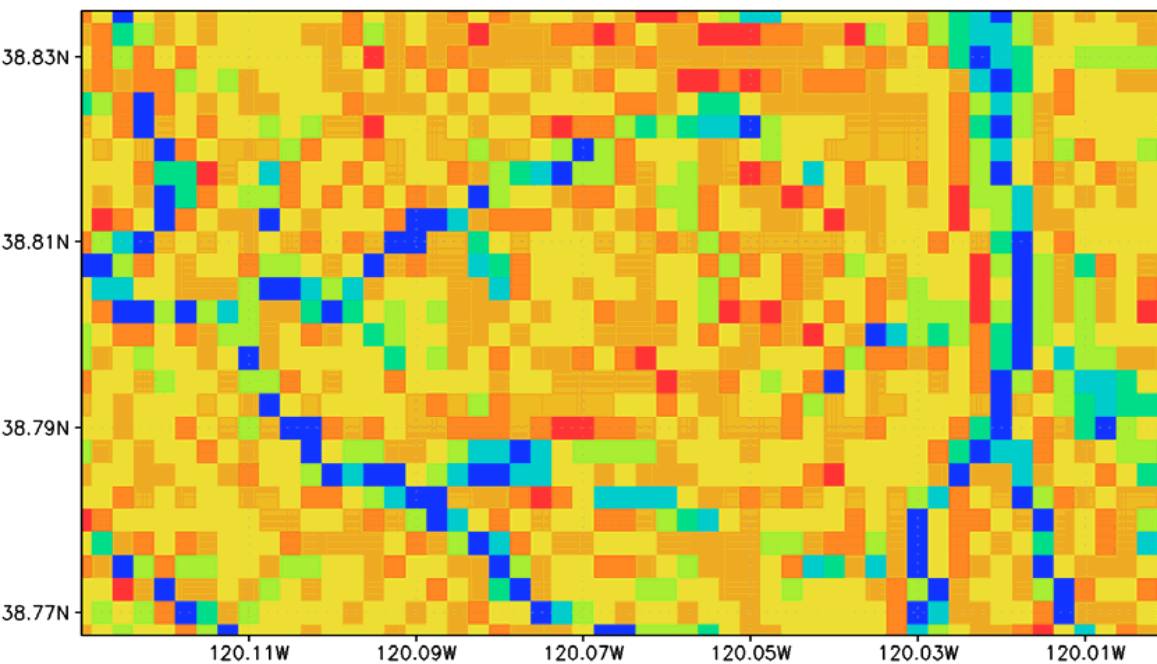
Advantages:

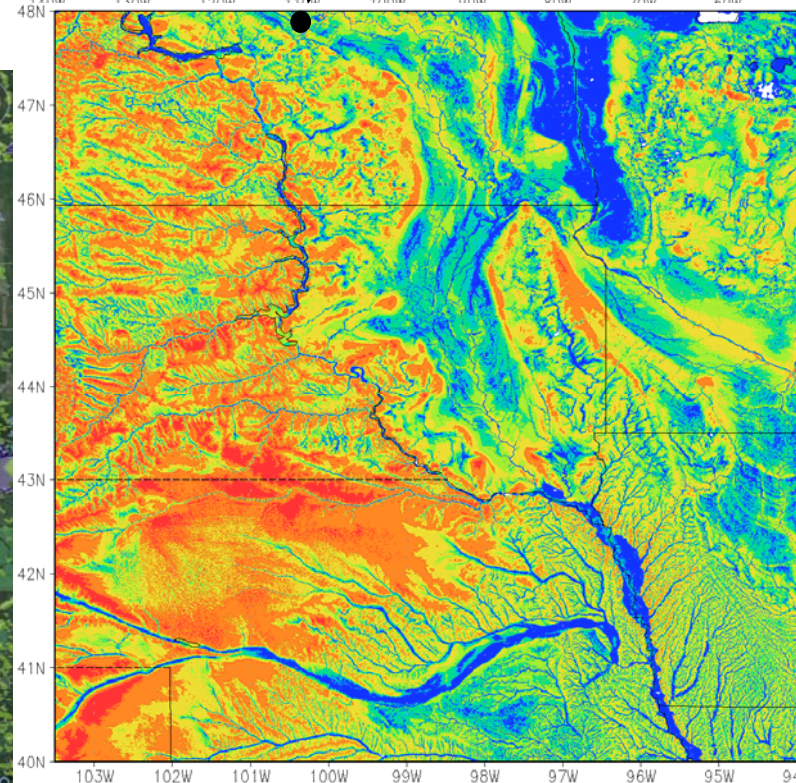
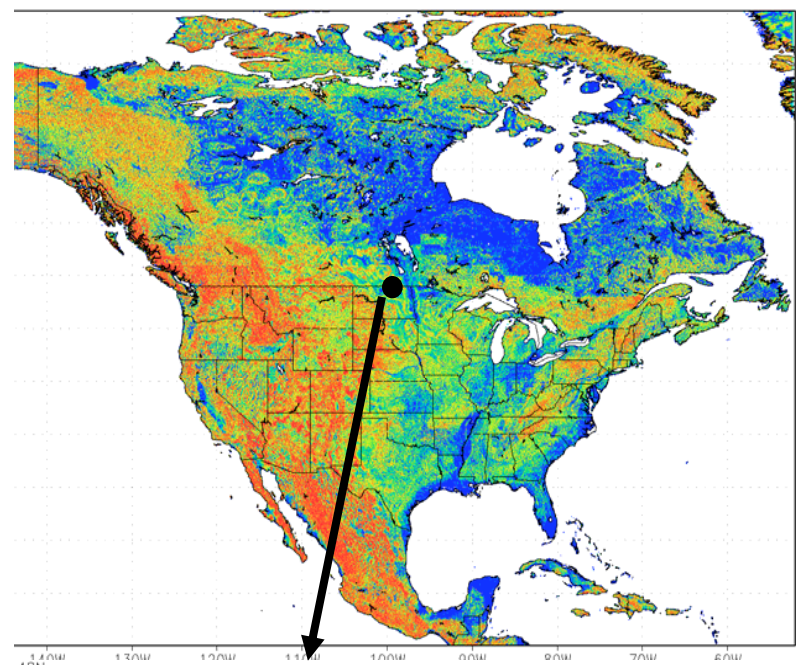
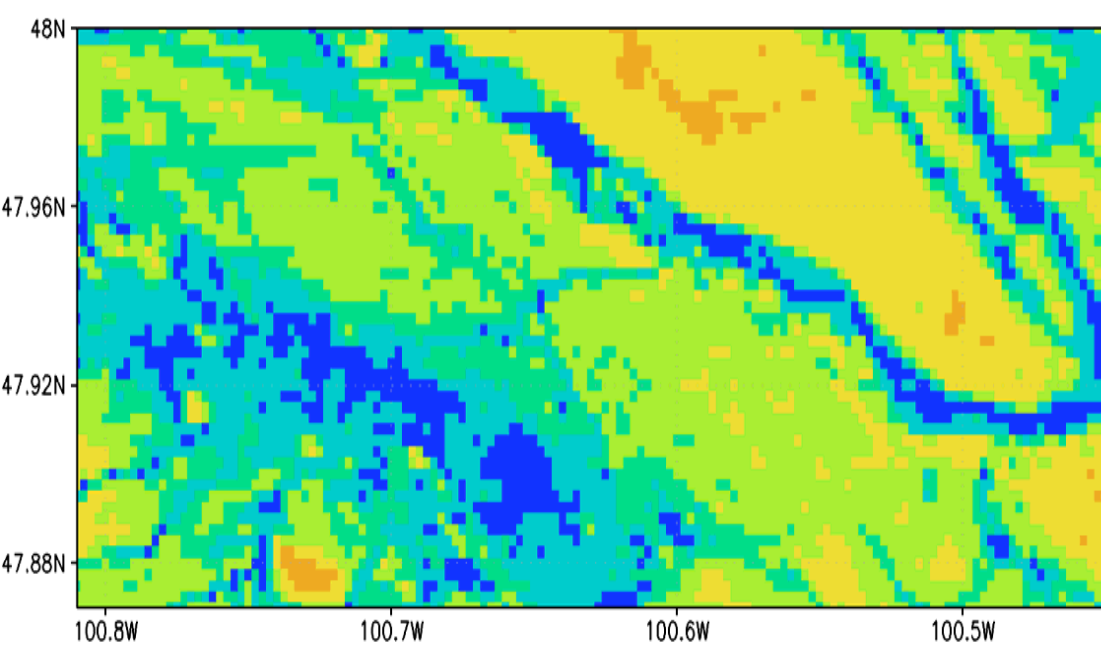
- Fully coupled, internally consistent
- Enabled several interfaces associated



Wetland Area
with Abundant
lands

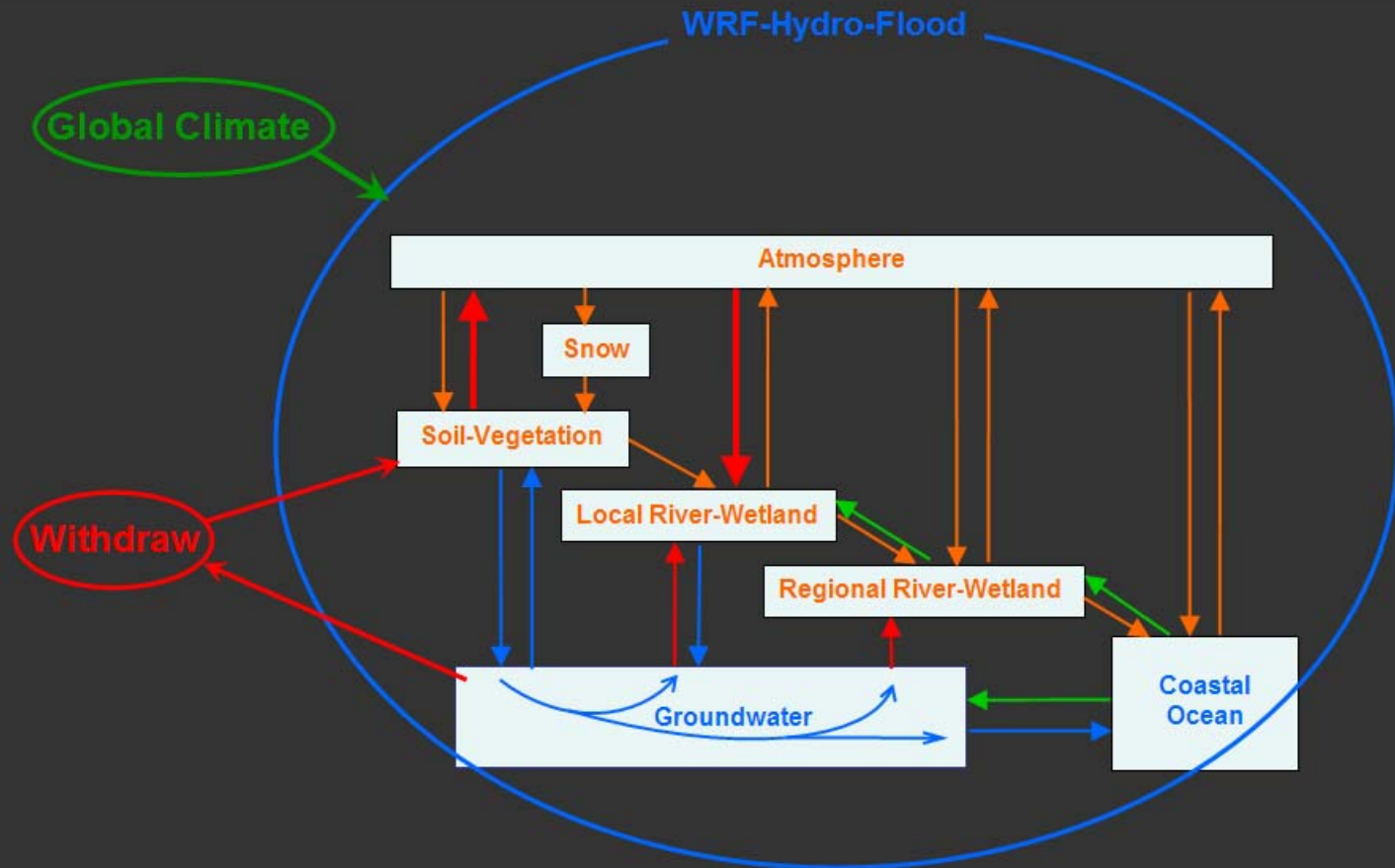






Project Tasks:

1. Reconstruct the hydrology over the past decades
2. Reconstruct the coupled climate-hydrology
3. Project the coupled climate-hydrology in mid and end of century



Project Tasks:

1. Reconstruct the hydrology over the past decades (1979-2011)

- Test and improve land hydrology

Space:

Lambert projection
5 km grids with $\rightarrow$ 1,450x1,510 (2.2)

Time:

2 min canopy, soil, flood routing
30 min water table, 1hr GW flow

Output: Daily

Soil moisture at 4 depths
intercept-evap-transpr fluxes
GW recharge, water table depth
GW-stream exchange
Stream flow, floodplain inundation



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- NCEP/NCAR Reanalysis

NCEP North American Regional Reanalysis: NARR

See the [PSD NARR webpage](#) for more information on PSD's involvement in the NARR project

Brief Description:

- NCEP's high resolution combined model and assimilated dataset. It covers 1979 to near present and is provided 8-times daily, daily and monthly on a Northern Hemisphere Lambert Conformal Conic grid for all variables. [More...](#)

Temporal Coverage:

- 8-times, Daily and Monthly means for 1979/01/01 to Dec 31, 2010.
- Long Term Daily, Monthly means for years 1979 - 2000.

Spatial Coverage:

- The native model grid is converted to a Northern Lambert Conformal Conic grid which is

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1. Reconstruct the hydrology over the past decades (1979-2011)

- Test and improve land hydrology

1. Flooding

Solving the full momentum equation

2. Obtaining Computation Resources

NCAR-CSL supercomputer (Bluefire)

Apr 2011 – June 2012

(2 days per model year)

NCAR-WSCN Peta-grid

June 2012-June 2014

3. Writing the code in parallel

Using 256 processors

4. Completed Model Setup

Land parameters

5. Compiling Validation Data

stream flow (78)

water table depth (813)

soil moisture (132)

6. Diagnostic Runs



Gonzalo Miguez-Macho

Hydrology over
1979-2011)

and hydrology

Deniz Kustu - Ph.D. Student



Haibin Li

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Supercomputer (Bluefire)

– June 2012

– 1 model year)

Peta-grid

– June 2014

– 1000 cores in parallel

– 100 processors

– 100 Model Setup

– 100 rs

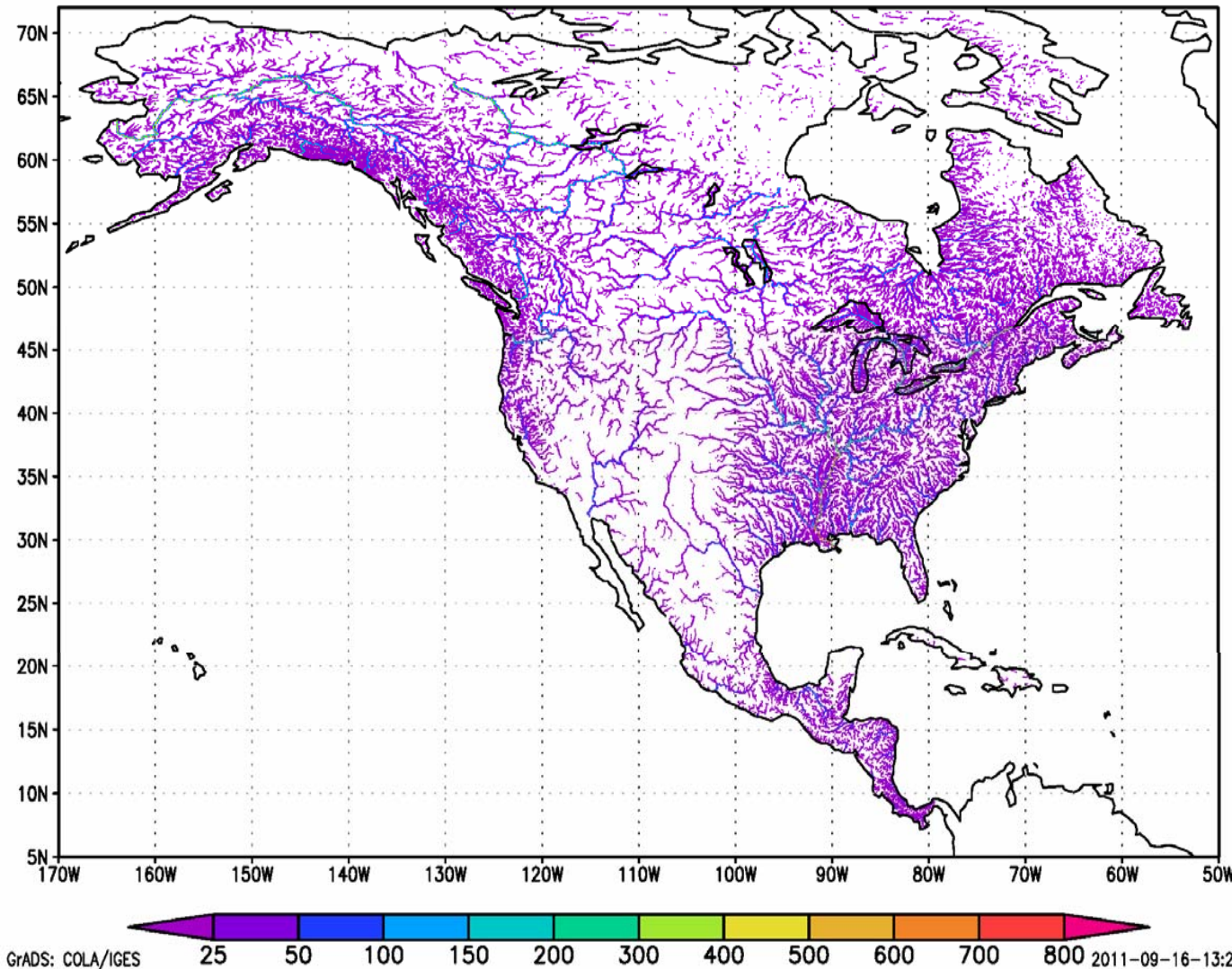
– 100 Relation Data

– 100)

– 100 th (813)

– 100 32)

– 100 S



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Solving the full momentum equation

2. Obtaining Computation Resources

USFWS, NSF, and DOE supercomputer (Bluefire)

– June 2012

– 1 model year)

Peta-grid

– June 2014

– 100,000 cores in parallel

– 100 processors

– Model Setup

– 100,000 cores

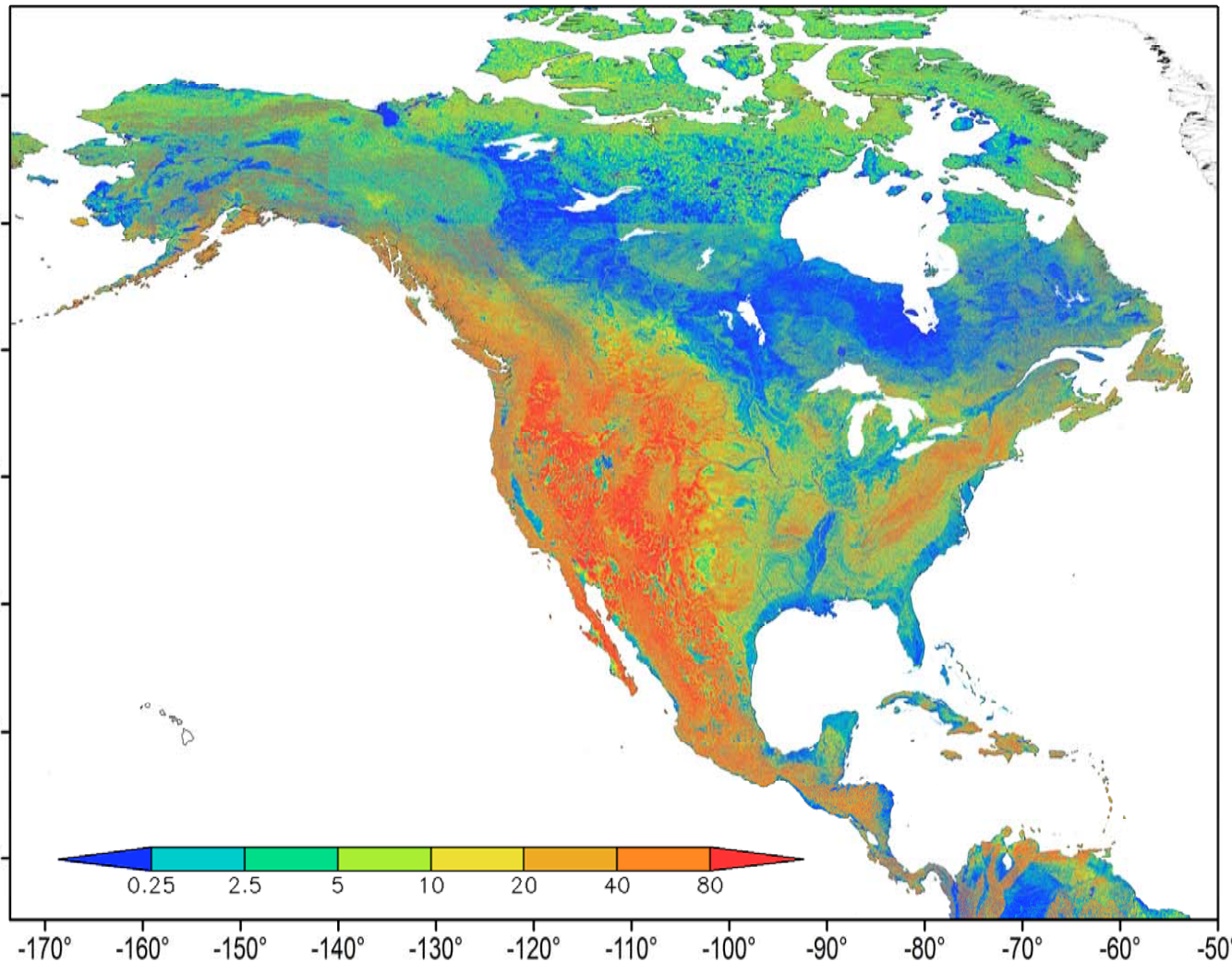
– Validation Data

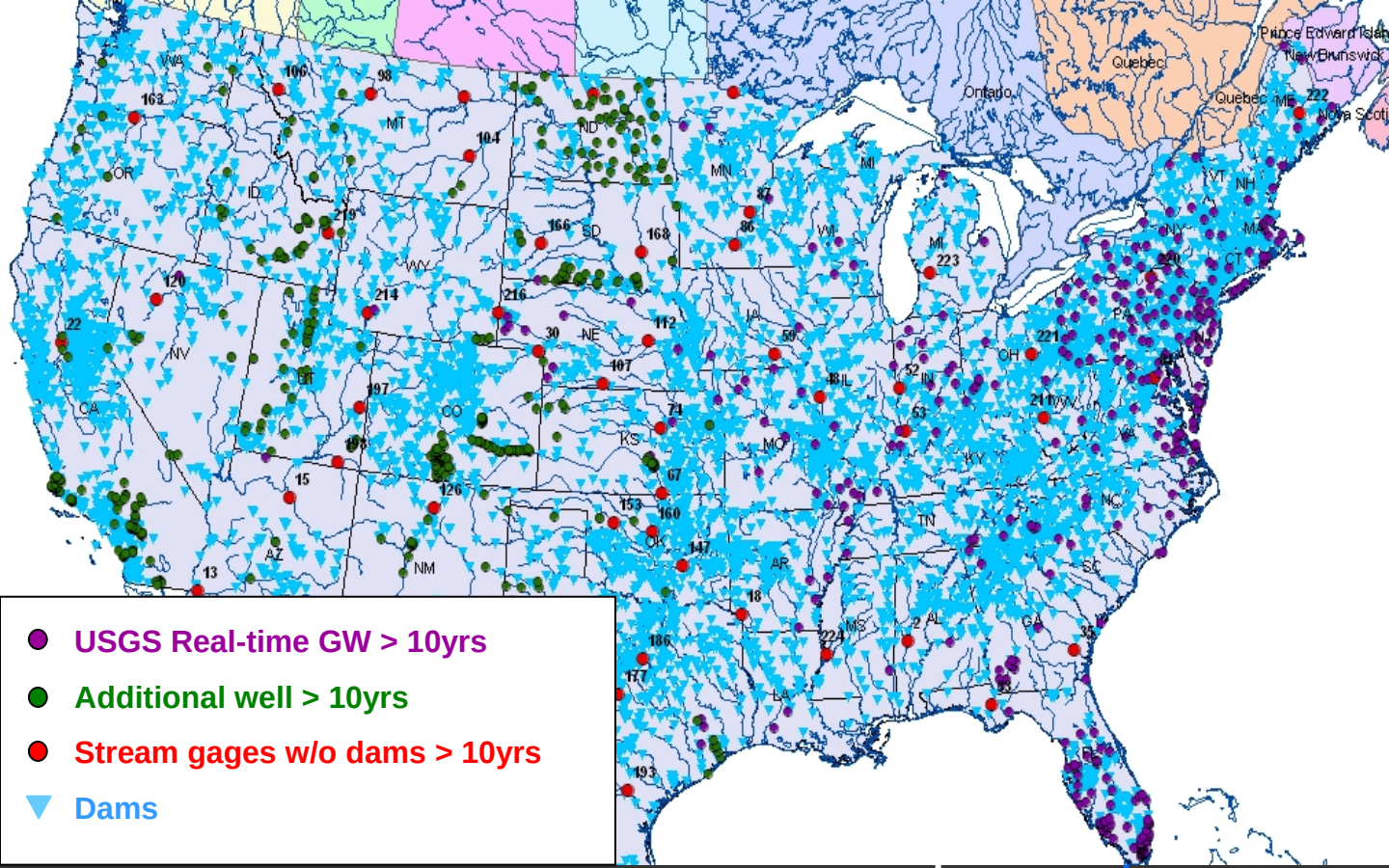
– 100,000 cores

– 100,000 cores (813)

– 100,000 cores (32)

– 100,000 cores





momentum equation

Validation Resources

computer (Bluefire)
 the 2012
 model year)

1-grid
 the 2014

parallel

ors

Setup

Land parameters

5. Compiling Validation Data

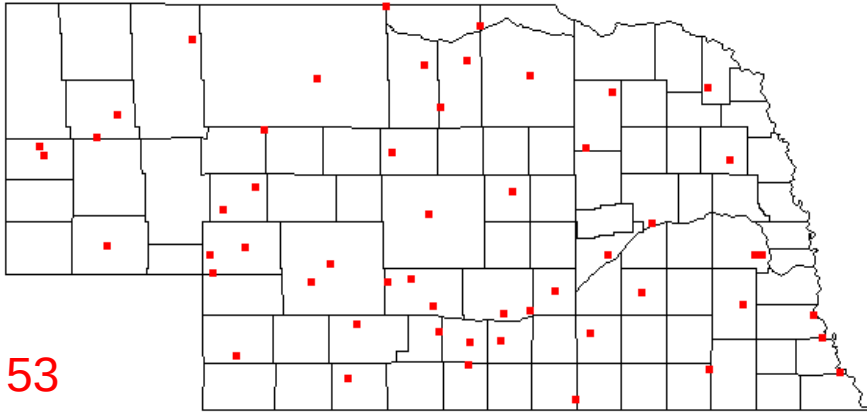
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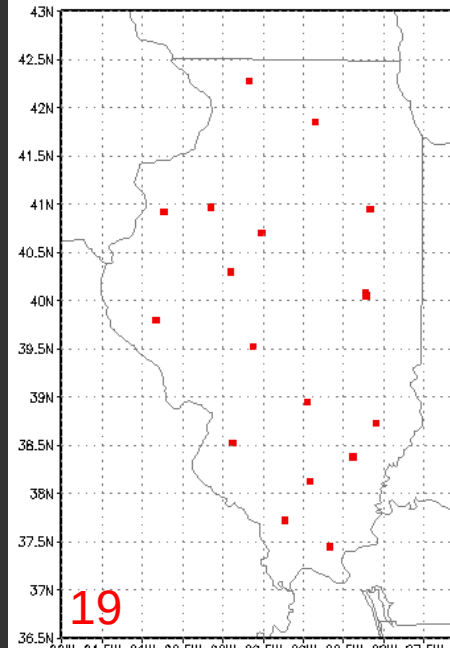
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Illinois Stations



momentum equation

Station Resources

computer (Bluefire)

June 2012

model year)

data-grid

June 2014

in parallel

Using 256 processors

4. Completed Model Setup

Land parameters

5. Compiling Validation Data

stream flow (78)

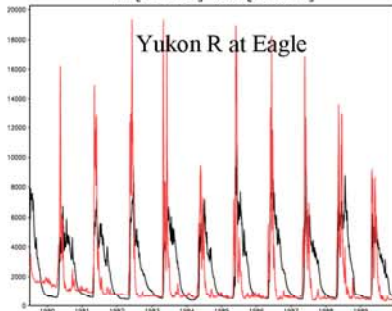
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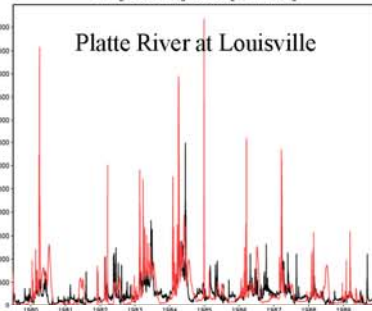
6. Diagnostic Runs

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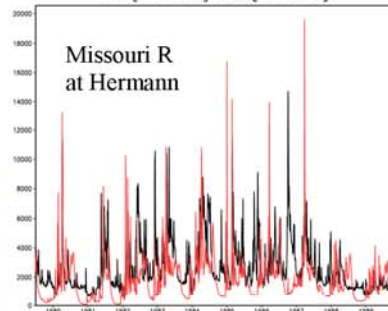
LH [286456.] USGS[293964.]



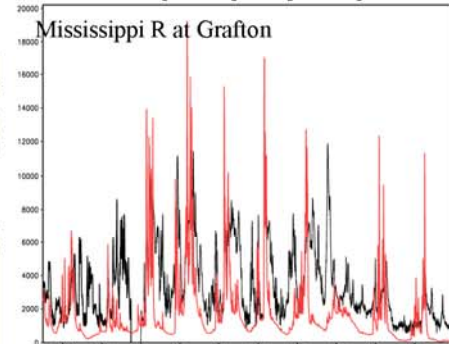
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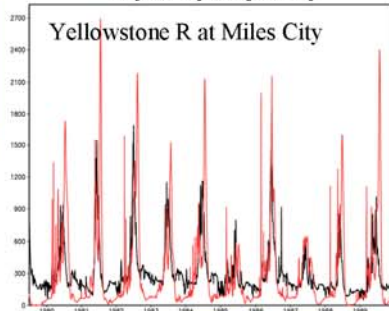
LH [1325360.] USGS[1353275.]



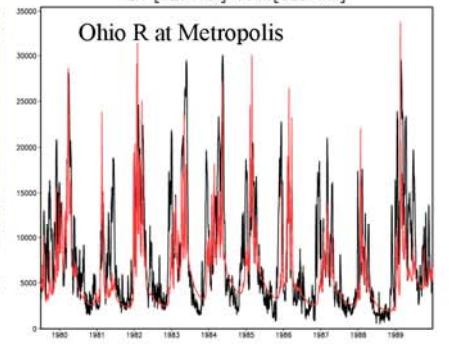
LH [444168.] USGS[443667.]



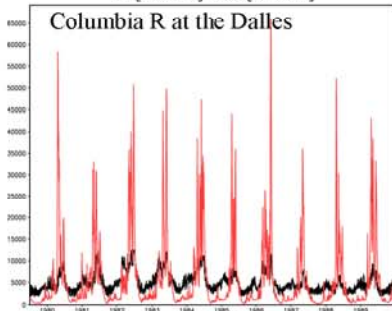
LH [124236.] USGS[124975.]



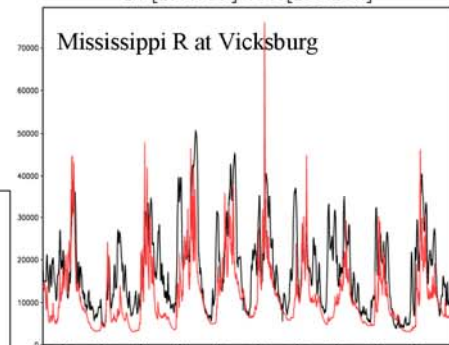
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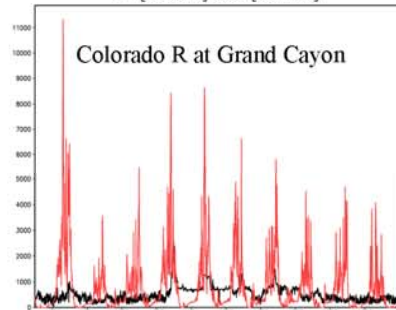
LH [549344.] USGS[613827.]



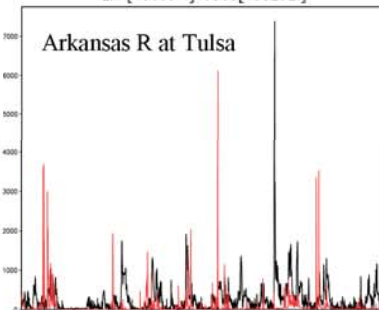
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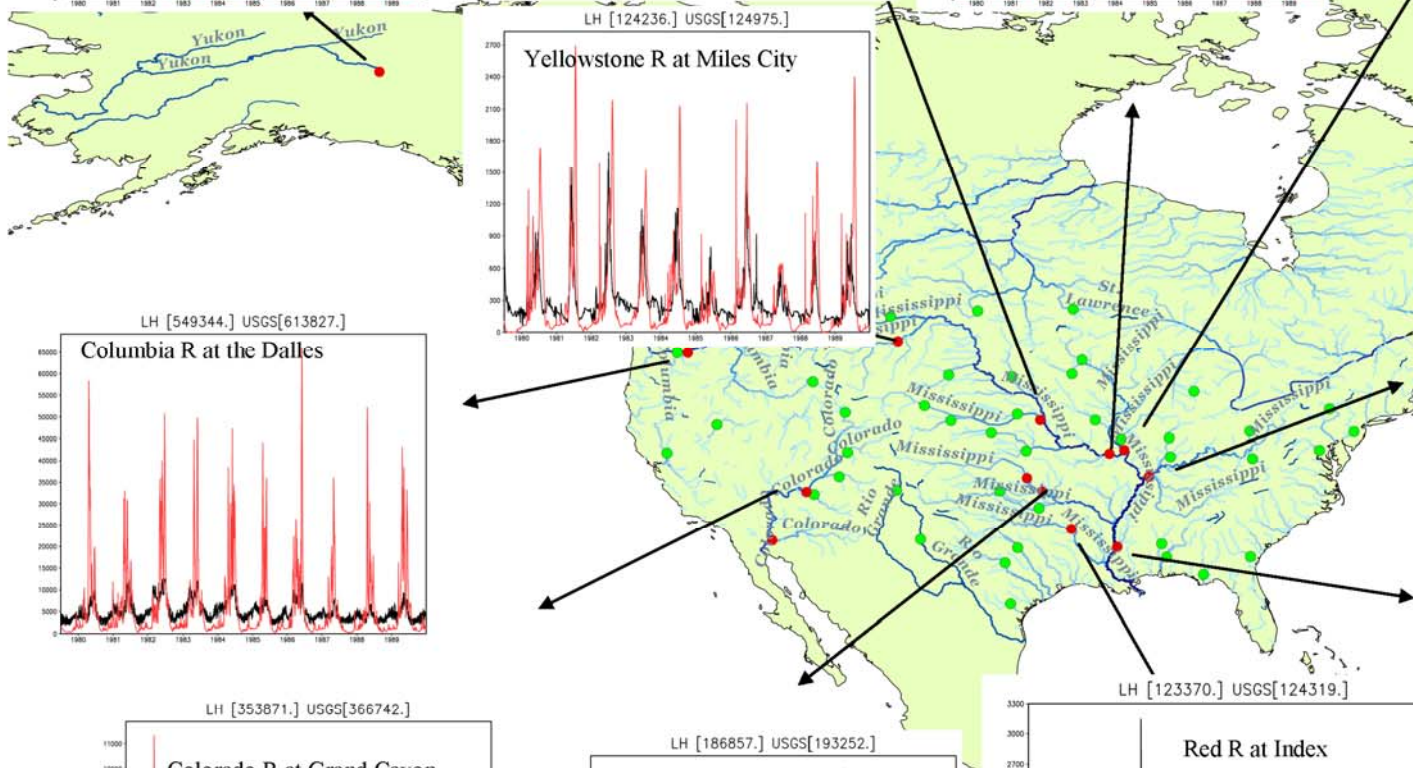
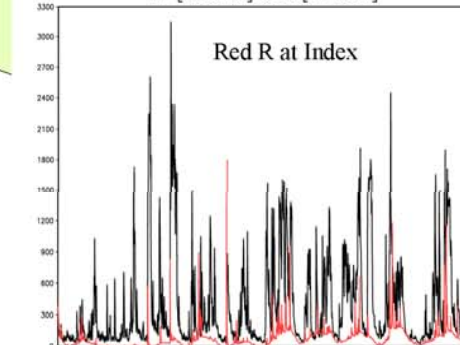
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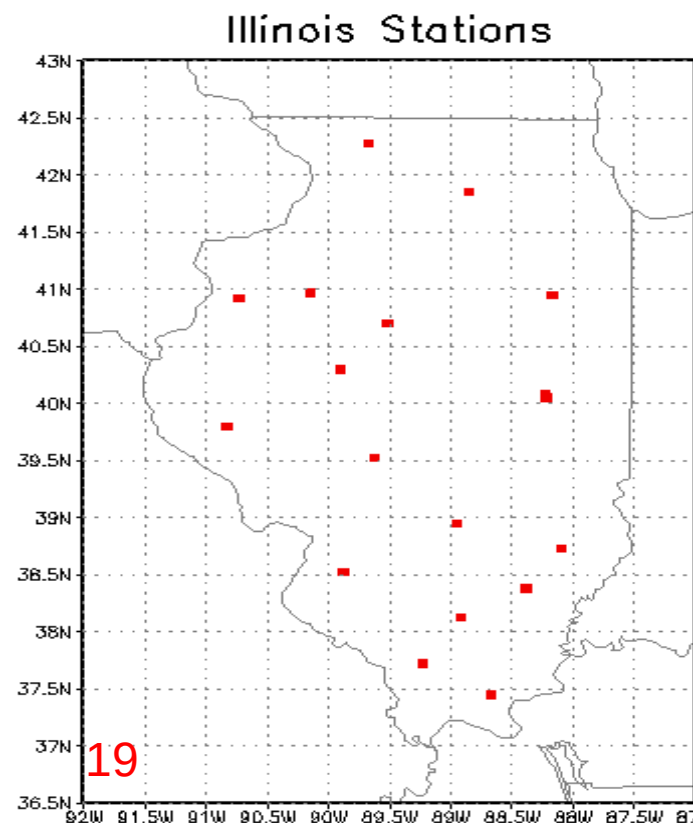
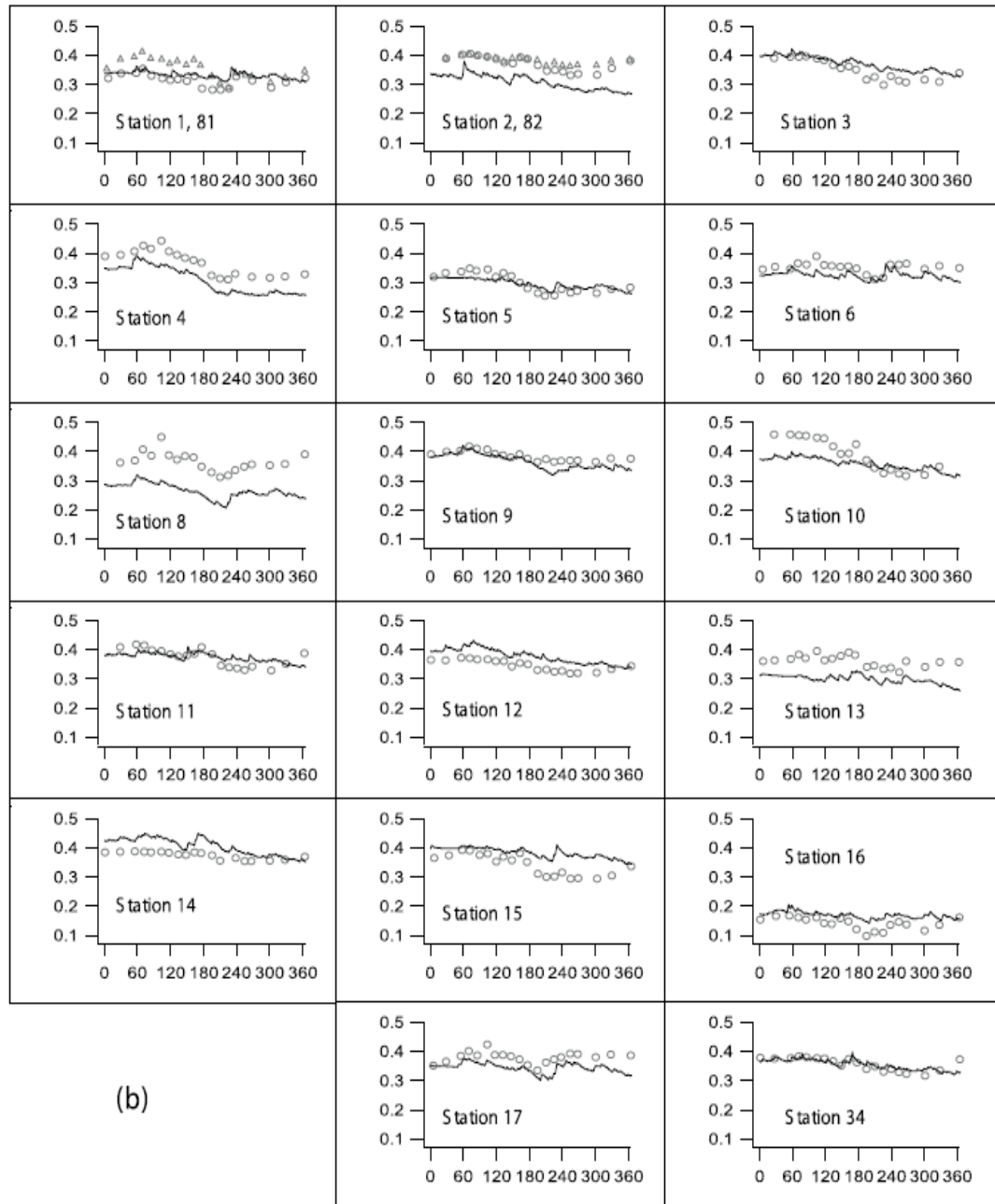


LH [186857.] USGS[193252.]



LH [123370.] USGS[124319.]





Compiling Validation Data

Stream flow (78)
 Water table depth (813)
 Soil moisture (132)

6. Diagnostic Runs

Top-1m Volumetric Soil Moisture



momentum equation

Simulation Resources

computer (Bluefire)

June 2012

model year)

data-grid

June 2014

in parallel

sors

Setup

Soil Moisture (%)

6. Diagnostic Runs

Continental-Scale Water Cycle Modeling:

Integrating Climate, Surface Water, Groundwater, Sea-Level, and Human Use

Project Goal:

- Assess US freshwater resources in the coming decades
- climate variability & change
- increasing human demands

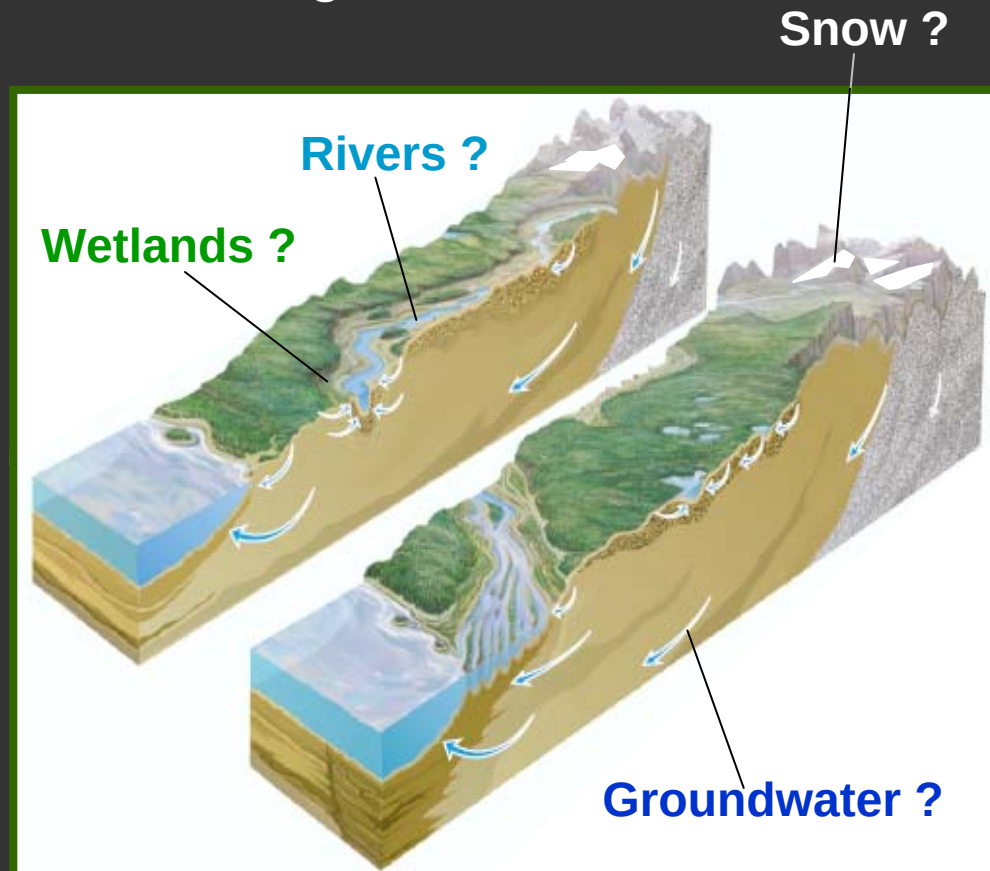
Water Quantity

Storage
Flow Regime
Residence Time
Pathways



Water Quality

Stream T
Sediment load
Stream Chem
Biota



- | | | | |
|----|---------------------------------|----|---|
| 1 | Northern Great Plains | 14 | Upper Colorado River Basin |
| 2 | High Plains | 15 | Oahu, Hawaii |
| 3 | Central Valley, California | 16 | Caribbean Islands |
| 4 | Northern Midwest | 17 | Columbia Plateau |
| 5 | Southwest alluvial basins | 18 | San Juan Basin |
| 6 | Florida | 19 | Michigan Basin |
| 7 | Northern Atlantic Coastal Plain | 20 | Edwards-Trinity |
| 8 | Southeastern Coastal Plain | 21 | Midwestern basins and arches |
| 9 | Snake River Plain | 22 | Appalachian valleys and Piedmont |
| 10 | Central Midwest | 23 | Puget-Willamette Lowland |
| 11 | Gulf Coastal Plain | 24 | Southern California alluvial basins |
| 12 | Great Basin | 25 | Northern Rocky Mountain Intermontane Basins |
| 13 | Northern glacial aquifers | | |

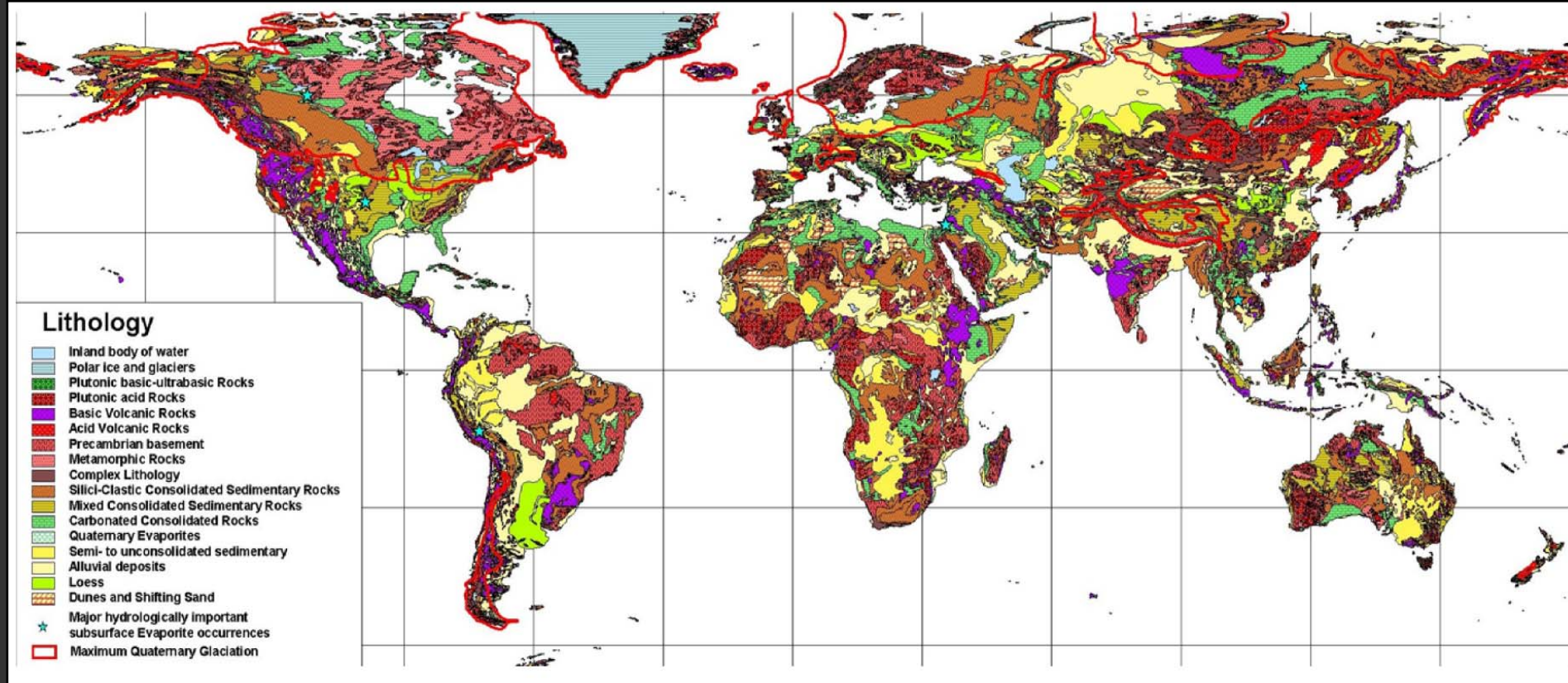
Lithologic composition of the Earth's continental surfaces derived from a new digital map emphasizing riverine material transfer

Hans H. Dürr and Michel Meybeck

UMR Sisyphe 7619, Université Pierre et Marie Curie/CNRS, Paris, France

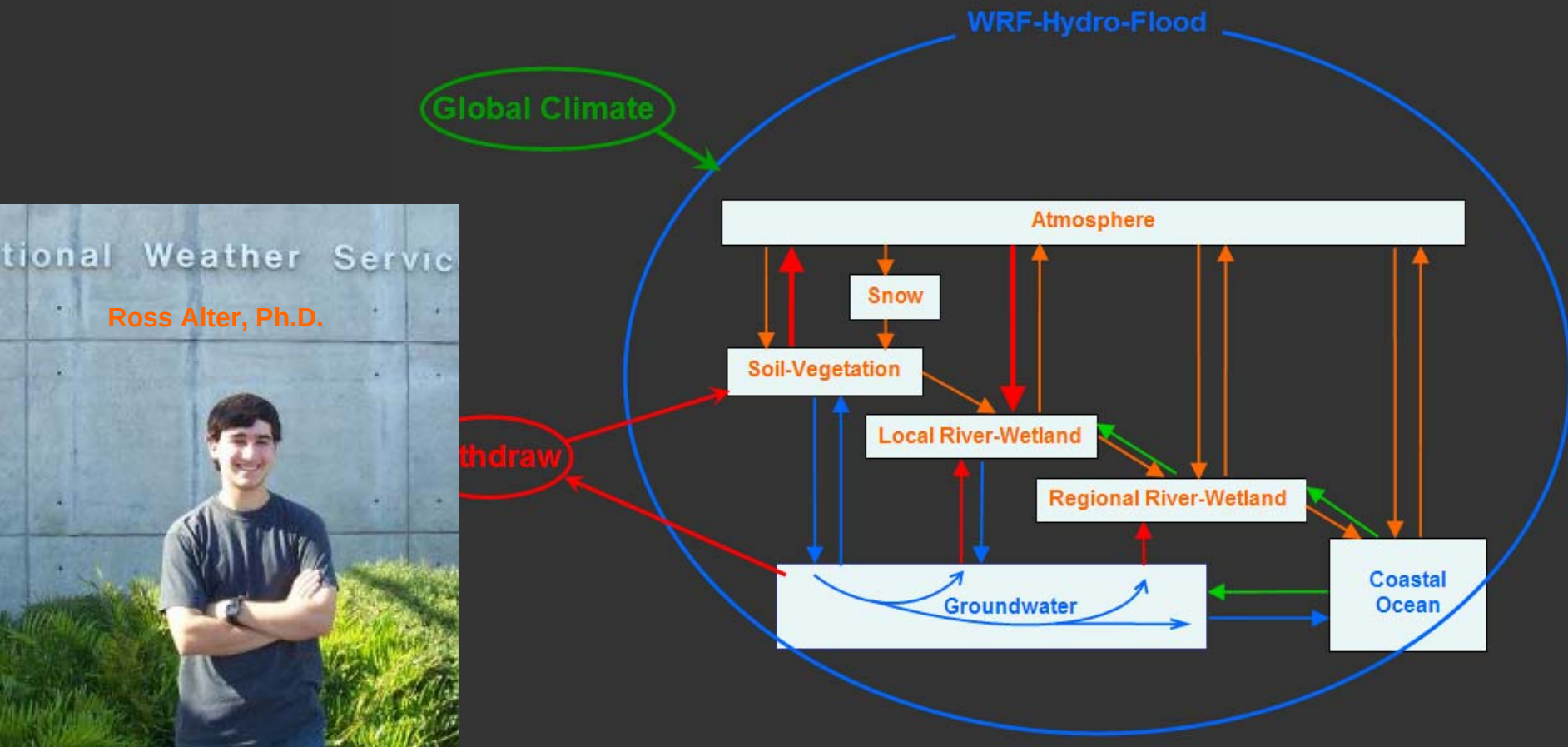
Stefan H. Dürr

Geologisches Institut, Universität Mainz, Mainz, Germany



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Funding:

EPA STAR

NSF water-carbon cycle

NSF climate & LS dynamics

NSF EaSM - NCAR

